BIOL 250 — Abdomen Comprehensive Study Notes

Ch 25 — The Digestive System

25.1 | An Overview of the Digestive System

- The digestive system = digestive tract (alimentary canal) + accessory organs. Digestive tract includes oral cavity, pharynx, esophagus, stomach, small intestine, large intestine. Accessory organs: teeth, tongue, salivary glands, liver, gallbladder, pancreas.
- Accessory glandular organs secrete water, enzymes, buffers, and other components into ducts that empty into the tract.
- Seven basic functions of digestive system:
  - **Ingestion**: food/liquids enter via mouth.
  - **Mechanical processing**: tongue/teeth before swallowing; churning, mixing, swirling, propulsive motions after.
  - **Digestion**: chemical/enzymatic breakdown of sugars, lipids, proteins into small molecules for absorption.
  - **Secretion**: acids, enzymes, buffers — some from tract lining, most from accessory organs (e.g., pancreas).
  - **Absorption**: movement of organic molecules, electrolytes, vitamins, water across epithelium into interstitial fluid.
  - **Excretion**: elimination of undigested residue and metabolic wastes (many wastes secreted by liver).
  - **Compaction**: progressive dehydration of undigested material/wastes -> feces. Defecation = elimination of feces via anus.
- Lining plays defensive role against: (1) corrosive digestive acids/enzymes, (2) mechanical stress/abrasion, (3) swallowed pathogens.

Histological Organization of the Digestive Tract (4 layers)

- **Mucosa**: inner lining; a mucous membrane = loose connective tissue covered by epithelium moistened by glandular secretions.
  - Epithelium is stratified (oral cavity, esophagus -- nonkeratinized stratified squamous, resists abrasion) or simple columnar (stomach, small intestine, most of large intestine -- specialized for secretion/absorption).
  - **Circular folds (plicae circulares)**: transverse/longitudinal mucosal folds that increase surface area for absorption/secretion; some permanent (mucosa+submucosa), some temporary (disappear as lumen fills).
  - **Lamina propria**: areolar connective tissue deep to epithelium; contains blood vessels, sensory nerve endings, lymphatics, smooth muscle fibers, scattered lymphatic tissue.
  - **Muscularis mucosae**: narrow band of smooth muscle + elastic fibers at mucosal border; two thin concentric layers (internal circular, external longitudinal); contraction alters lumen shape and moves epithelial folds.
- **Submucosa**: areolar connective tissue deep to muscularis mucosae. Contains large blood/lymphatic vessels; in some regions contains exocrine glands secreting buffers/enzymes into lumen. Contains **submucous neural plexuses** (sensory neurons, parasympathetic ganglia, sympathetic postganglionic fibers) that innervate the mucosa.
- **Muscular layer**: double layer of smooth muscle deep to submucosa -- internal circular and external longitudinal layers; mechanically process/propel materials. Movements coordinated by **myenteric neural plexus** (between the two muscle layers) = parasympathetic ganglia + sympathetic postganglionic fibers.

Parasympathetic stimulation increases muscular tone/contractions; sympathetic decreases tone/promotes relaxation.

  - Localized thickenings of circular layer form **sphincters (valves)** -- constrict to restrict movement or ensure one-way passage.
- **Serosa**: serous membrane covering the muscular layer within the peritoneal cavity. Absent around pharynx, esophagus, and rectum -- instead these are wrapped by a fibrous sheath of collagen called the **adventitia**.

Muscular Layers and Movement of Digestive Materials

- Smooth muscle cells: 5-10 um diameter, 30-200 um length; surrounded by connective tissue but (unlike skeletal muscle) collagen fibers don't form tendons/aponeuroses; contractile proteins not organized into sarcomeres -> smooth muscle cells lack striations, yet contract as strongly as skeletal/cardiac muscle.
- **Plasticity**: ability of stretched smooth muscle to adapt to new length and still contract on demand -- important for organs with large volume changes (e.g., stomach).
- Smooth muscle cells are involuntary; many lack direct ANS motor innervation.
- **Pacesetter cells**: located in muscularis mucosae and muscular layer; undergo spontaneous depolarization triggering contraction. **Gap junctions** electrically connect adjacent muscle cells so one stimulated cell can trigger contraction of a whole region (functional syncytium); may also respond to hormones, chemicals, O2/CO2 levels, stretching, or irritation.
- **Peristalsis**: contractions of muscularis mucosae/muscular layer that move a **bolus** (small oval mass of food) along the tract. Circular muscles contract behind the bolus, then longitudinal muscles contract (shortening the segment), producing a wave that forces contents forward.
- **Segmentation**: occurs in most of small intestine and some large intestine regions; churns/fragments digestive materials, mixing with secretions; does NOT produce net directional movement.
- Peristalsis/segmentation also triggered by hormones, chemicals, physical stimulation. Afferent/efferent fibers of glossopharyngeal, vagus, or pelvic nerves initiate peristaltic waves.
- **Short (myenteric) reflexes**: local, sensory receptors in tract wall -> myenteric plexus synapse -> localized peristalsis over a few cm, no CNS involvement. **Enteric nervous system** = neural network coordinating these reflexes; has roughly as many neurons/neurotransmitters as the spinal cord.
- **Long reflexes**: sensory info also travels to CNS, involving interneurons/motor neurons for higher-level, larger-scale control (motor fibers via glossopharyngeal, vagus, or pelvic nerves).

The Peritoneum

- Serous membrane with two parts: **visceral peritoneum** (serosa, covers organs) continuous with **parietal peritoneum** (lines body wall). Continually secretes watery lubricating fluid -- about **7 liters/day secreted and reabsorbed**, but normal peritoneal cavity volume at any time is very small (~100-400 mL).
- Organ-peritoneum relationships:
  - **Intraperitoneal organs**: lie within peritoneal cavity, covered on all sides by visceral peritoneum (e.g., stomach, liver, ileum).
  - **Retroperitoneal organs**: covered by parietal peritoneum on anterior surface only; lie outside peritoneal cavity; typically do NOT develop from embryonic gut (e.g., kidneys, ureters, abdominal aorta).
  - **Secondarily retroperitoneal organs**: form as intraperitoneal organs during development but become retroperitoneal as visceral peritoneum fuses with opposing parietal peritoneum (e.g., pancreas, distal two-thirds of duodenum).
- **Mesenteries**: fused double sheets of peritoneal membrane connecting parietal to visceral peritoneum; areolar connective tissue between the two surfaces carries blood vessels, nerves, lymphatics; stabilize organ position, prevent entanglement.
  - Ventral mesentery mostly disappears during development except: (1) **lesser omentum** (stomach to liver), and (2) **falciform ligament** (liver to anterior abdominal wall/diaphragm). (Note: though called "ligament," not comparable to bone ligaments.)
  - **Greater omentum**: dorsal mesentery of stomach enlarges into a pouch extending inferiorly between body wall and anterior small intestine surface; contains thick adipose layer (energy reserve + insulation

  - minimizing heat loss across abdominal wall) and numerous lymph nodes (protect against foreign proteins/toxins/pathogens).
  - **Mesentery proper**: thick mesentery suspending all but the first 25 cm of the small intestine; provides stability + independent movement.
  - **Mesocolon**: mesentery attached to large intestine. **Transverse mesocolon** suspends transverse colon; **sigmoid mesocolon** suspends sigmoid colon.
  - During development, dorsal mesentery of ascending colon, descending colon, and rectum fuses to posterior body wall, fixing them as secondarily retroperitoneal (visceral peritoneum covers only anterior/part of lateral surfaces).
- **Clinical: Ascites** -- buildup of serous fluid in peritoneal cavity beyond normal 100-400 mL, causing abdominal swelling, discomfort, heartburn, indigestion, shortness of breath, lower back pain. Most common cause = advanced liver disease or hepatitis B/C infection; also kidney failure, heart failure, advanced cancer. Treatment: sodium restriction + diuretics.

25.2 | The Oral Cavity

- Functions of oral cavity: (1) sensory analysis of food before swallowing, (2) mechanical digestion via teeth/tongue/palate, (3) lubrication via mucus and saliva, (4) limited chemical digestion of carbohydrates (salivary amylase) and lipids (lingual lipase).
- **Oral mucosa**: nonkeratinized stratified squamous epithelium protecting mouth from abrasion.
- **Cheeks**: lateral walls of oral cavity, supported by buccal fat pads and buccinator muscle; anteriorly continuous with lips.
- **Oral vestibule**: space between cheeks/lips and teeth.
- **Gingivae** (gums): ridges of oral mucosa surrounding tooth bases on alveolar processes of maxillae / alveolar part of mandible.
- **Tongue**: forms floor of oral cavity; mylohyoid (inferior to tongue) supports oral floor. Four functions: (1) mechanical digestion via compression/abrasion/distortion, (2) manipulation to assist chewing/prepare bolus for swallowing, (3) sensory analysis (touch, temperature, taste), (4) secretion of mucin and lingual lipase.
  - Divided into anterior **body** and posterior **root**.
  - Dorsum has **lingual papillae** (fine projections; thick stratified squamous epithelium for friction); taste buds along edges of many papillae.
  - **Circumvallate papillae**: V-shaped line marking boundary between body and root.
  - Epithelium flushed by water, mucin, lingual lipase from small glands in lamina propria; **lingual lipase** begins enzymatic breakdown of triglycerides.
  - Inferior surface epithelium thinner/more delicate; **frenulum of the tongue** (midline fold) connects tongue body to floor mucosa, limits extreme movement. Overly restrictive frenulum -> **ankyloglossia** ("tongue-tied"), correctable surgically.
  - Two muscle groups: **extrinsic** (hyoglossus, styloglossus, genioglossus, palatoglossus -- gross movements) and **intrinsic** (change tongue shape, assist precise movements like speech); both controlled by **hypoglossal nerve (CN XII)**.
- **Hard palate**: formed by palatine processes of maxilla + horizontal plates of palatine bones; separates oral cavity from nasal cavity.
- **Soft palate**: posterior to hard palate; separates oral cavity from nasopharynx, closes off nasopharynx during swallowing.
- **Uvula**: finger-shaped projection from posterior margin of soft palate; prevents food from entering pharynx too soon.
- **Palatal arches** (2 pairs flanking uvula):
  - **Palatoglossal arches**: soft palate to tongue base (palatoglossus muscle).
  - **Palatopharyngeal arches**: soft palate to pharynx side (palatopharyngeus muscle).
- **Palatine tonsils**: lie between palatoglossal and palatopharyngeal arches.

- **Fauces**: space between oral cavity and pharynx, bounded by soft palate and tongue base.
- **Salivary glands** (3 pairs) secrete into oral cavity:

1. **Parotid glands**: largest (~20 g each); between zygomatic arch and sternocleidomastoid anteriorly, from mastoid process across masseter posteriorly; drain via **parotid duct** into oral vestibule opposite second upper molar. Produce serous, enzyme-rich saliva (dominated by serous secretory cells).

2. **Sublingual glands**: in mucous membrane of mouth floor; numerous **sublingual ducts** open along frenulum of tongue. Mucin-rich saliva (mucous secretory cells dominant).

3. **Submandibular glands**: floor of mouth along medial mandible, inferior to mylohyoid line; **submandibular ducts** open beside frenulum, posterior to teeth. Mixed serous+mucous secretions (enzymes + mucin).

- **Saliva composition/production**: ~70% from submandibular glands, 25% from parotid, 5% from sublingual. ~1.0-1.5 L/day produced; 99.4% water, remainder ions/buffers/metabolites/enzymes. **Mucin** glycoproteins give lubricating action.
- **Regulation of salivary secretion**: ANS control (parasympathetic + sympathetic innervation to each gland). Food triggers salivary reflex via trigeminal nerve receptors or taste buds (CN VII, IX, X). Parasympathetic stimulation -> large volume of saliva. Sympathetic role unclear -- evidence suggests small amounts of thick saliva.
- **Clinical: Mumps** -- mumps virus infection, most common ages 5-9; spread person-to-person via respiratory droplets/saliva. Targets salivary glands (usually parotid) -> swollen, painful glands. In post-adolescent males may infect testes (-> sterility) or pancreas (-> temporary/permanent diabetes). Severe cases may affect CNS. First exposure -> lasting immunity. Vaccine has nearly eliminated disease in U.S. but persists in developing countries.

The Teeth

- **Mastication** (chewing): breaks down connective tissues in meat/plant fibers, saturates food with saliva/enzymes.
- Tooth structure: **dentine** (mineralized matrix like bone but without living cells; cells in pulp cavity extend processes into it); **pulp cavity** (spongy, highly vascular; contains blood vessels/nerves entering via **root canal** at tooth base); **apical foramen** at root tip -- dental artery, dental vein, dental nerve enter here.
- **Tooth socket (alveolus)**: bony cavity holding the root; **periodontal ligament** (collagen fibers from dentine of root to bone) creates a **gomphosis** joint.
- **Cement**: covers root dentine, histologically similar to bone, less resistant to erosion than dentine; protects/anchors periodontal ligament.
- **Neck**: boundary between root and **crown** (visible portion above gingiva). **Gingival sulcus**: shallow groove around tooth neck, prevents bacterial access to lamina propria/cement; breakdown -> **gingivitis** (gum inflammation).
- **Enamel**: covers crown dentine, forms **occlusal surface** (biting surface); contains densely packed calcium phosphate crystals -- hardest biologically manufactured substance. Adequate calcium, phosphate, vitamin D needed in childhood for complete/decay-resistant enamel. **Cusps** = elevations of occlusal surface.
- **Types of teeth** (upper/lower **dental arcades** on alveolar processes of maxillae/mandible):

1. **Incisors**: blade-shaped, front of mouth; clip/cut food; single root.

2. **Canines (cuspids)**: conical, sharp ridgeline/pointed tip; tear/slash food; single root.

3. **Premolars (bicuspids)**: flattened crowns, two rounded cusps; crush/mash/grind; one or two roots.

4. **Molars**: large flattened crowns, four to five cusps; crush/grind; upper molars typically 3 roots, lower usually 2 roots.

- **Sets of teeth**: **Deciduous** ("baby") teeth appear first -- 20 total (per side: 2 incisors, 1 canine, 2 deciduous molars). Replaced by **permanent** teeth of larger adult jaws as periodontal ligaments/roots of deciduous teeth erode; adult premolars replace deciduous molars, adult molars extend rows as jaw grows. Three additional molars appear per side with age -> permanent count = **32**.
- **Third molars (wisdom teeth)**: may not erupt before age 21; may fail to erupt due to malposition or insufficient space.

- **Mastication mechanics**: masticatory muscles close/slide/rock the jaw; buccal, labial, lingual muscles control food position across occlusal surfaces; tongue compacts food into a swallowable bolus.
- **Clinical**: **Periodontal disease** -- loosening of teeth in alveolar sockets from periodontal ligament erosion by bacterial acids.

25.3 | The Pharynx

- Common passageway for food, liquids, and air; divisions (from Ch 24): nasopharynx, oropharynx, laryngopharynx. Deep to lamina propria of mucosa is a dense elastic fiber layer bound to underlying skeletal muscles.
- Pharyngeal muscles involved in swallowing (from Ch 10):
  - **Superior, middle, inferior pharyngeal constrictors**: push bolus toward esophagus.
  - **Palatopharyngeus and stylopharyngeus**: elevate the larynx.
  - **Palatal muscles**: raise soft palate and adjacent pharyngeal wall.
- **Deglutition (swallowing)** -- three phases:

1. **Buccal phase** (voluntary): bolus compressed against hard palate; tongue retracts -> forces bolus into pharynx, helps elevate soft palate, isolates nasopharynx.

2. **Pharyngeal phase** (involuntary): begins when bolus contacts palatal arches/posterior pharyngeal wall. Palatopharyngeus/stylopharyngeus elevate larynx; epiglottis folds to direct bolus past closed glottis; pharyngeal constrictors propel bolus into esophagus in <1 second; respiratory centers inhibited, breathing ceases.

3. **Esophageal phase**: begins when upper esophageal sphincter opens; peristaltic waves push bolus through esophagus; lower esophageal sphincter opens as bolus approaches stomach.

25.4 | The Esophagus

- Hollow muscular tube transporting food/liquids to stomach; lies posterior to trachea, slightly left of midline; travels along dorsal mediastinal wall in thorax; enters peritoneal cavity through the **esophageal hiatus** in the diaphragm.
- Begins at level of **cricoid cartilage**, anterior to vertebra **C6**; ends anterior to vertebra **T7**.
- **Upper esophageal sphincter** and **lower esophageal sphincter (cardiac sphincter)**: functional sphincters (not distinct anatomical structures) named for functional similarity to true sphincters elsewhere.
- **Blood supply**: esophageal arteries, thyrocervical trunk, external carotid arteries (neck); bronchial arteries and esophageal arteries (mediastinum); inferior phrenic artery and left gastric artery (abdomen). Venous drainage: esophageal, inferior thyroid, azygos, and gastric veins.
- Innervation: vagus and sympathetic trunks via the **esophageal plexus**.

Histology of the Esophageal Wall

- Same mucosa/submucosa/muscular layers as elsewhere, with distinctive features:
  - Mucosal epithelium: abrasion-resistant nonkeratinized stratified squamous.
  - Mucosa + submucosa form large longitudinal folds allowing expansion for a bolus; muscle tone keeps esophagus closed except during swallowing.
  - Muscularis mucosae: very thin/absent near pharynx, thickens to **200-400 um** near stomach junction; composed of a single longitudinal smooth muscle layer only (differs from rest of tract).
  - Submucosa contains **esophageal glands** (simple branched tubular; mucous secretion lubricates bolus, protects epithelium) -- esophagus is one of only two tract regions with submucosal glands (the other is the **duodenum**).
  - Muscular layer: inner circular + outer longitudinal. Superior third = mostly skeletal muscle + some smooth; middle third = even mix skeletal/smooth; inferior third = only smooth muscle. Involuntary visceral reflexes control both skeletal and smooth portions.
  - **Adventitia** anchors esophagus to posterior body wall. Between diaphragm and stomach, esophagus is retroperitoneal; peritoneum covers only anterior and left lateral surfaces here.

- **Clinical: Esophagitis** -- inflammation of stratified squamous epithelium; most common cause = lower esophageal sphincter failure allowing acid reflux ("heartburn," resembling cardiac pain). Chronic recurrent reflux -> **GERD** (gastroesophageal reflux disease). Inhaling acidic contents can cause asthma. Worsened by obesity, pregnancy, aging, citrus, tomato products, strong seasoning. Treatment: healthy weight, smoking cessation, dietary change, smaller meals, remaining upright >=2 hrs post-meal.
- **Clinical: Hiatal hernia** -- stomach/other abdominal organs push upward through weakened esophageal hiatus; common in elderly, not always symptomatic, can contribute to GERD.

25.5 | The Stomach

- Functions: (1) temporarily store ingested food, (2) mechanically digest food, (3) chemically digest food via acids/enzymes -> produces **chyme** (viscous, strongly acidic, soupy mixture).
- Intraperitoneal, J-shaped; extends between vertebral levels **T7-L3**.
- **Four regions**:

1. **Cardia**: superior/medial portion surrounding esophageal junction; esophageal lumen opens at the **cardiac orifice**.

2. **Fundus**: projects superior to esophagus-stomach junction; contacts inferior/posterior diaphragm.

3. **Body**: between fundus and curve of the J; largest region; functions as mixing tank for food + gastric secretions.

4. **Pyloric part**: between body and duodenum; subdivided into **pyloric antrum** (connects to body), **pyloric canal** (empties into duodenum), and **pylorus** (muscular tissue around pyloric orifice/stomach outlet). **Pyloric sphincter** = thickened circular muscle regulating chyme release into duodenum.

- **Gastric folds (rugae)**: appear when stomach is empty, flatten as stomach distends (volume increases at meals, decreases as chyme empties).
- **Blood supply** -- three branches of celiac trunk:

1. **Left gastric artery**: lesser curvature and cardia.

2. **Splenic artery**: fundus directly, greater curvature via **left gastro-epiploic artery**.

3. **Common hepatic artery** -> **right gastric**, **right gastro-epiploic**, **gastroduodenal arteries**: greater/lesser curvatures at pylorus.

    - Venous drainage: gastric and gastro-epiploic veins -> **hepatic portal vein**.

- **Mesenteries of stomach (omenta)**: **lesser omentum** (hepatogastric ligament: liver to lesser curvature; hepatoduodenal ligament: liver to duodenum, contains bile duct/hepatic artery/portal vein en route to porta hepatis); **greater omentum** (hangs like apron from greater curvature).
- Clinical device note: **Endoscope** -- flexible fiberoptic tube w/ light+camera for direct visualization; **EGD (esophagogastroduodenoscopy)** = upper GI endoscopy via mouth; **colonoscopy** = endoscope via anus to examine colon.

Musculature of the Stomach

- Muscularis mucosae has an extra **outer circular layer**; muscular layer has an extra **inner oblique layer** in addition to usual circular + longitudinal -- these extra layers strengthen the wall and aid mixing/churning to form chyme.

Histology of the Stomach

- Simple columnar epithelium lines all regions; **mucous surface cells** form a protective mucus layer over the lumen.
- **Gastric pits**: shallow depressions on gastric surface; **regenerative stem cells** at base continuously replace shed superficial cells (defense mechanism).
- Each gastric pit connects to several **gastric glands** (simple branched tubular) extending into lamina propria, containing 4 secretory cell types. Parietal + chief cells together secrete ~**1500 mL of gastric juice/day**.
    - **Mucous neck cells**: found in all stomach regions; columnar; produce water-soluble lubricating secretion.

    - **Parietal cells**: large, round/pyramid-shaped; numerous in proximal gastric glands. Secrete **intrinsic factor** (glycoprotein aiding vitamin B12 absorption in intestine) and **hydrochloric acid (HCl)** -- lowers gastric juice pH, kills microorganisms, breaks down plant cell walls/meat connective tissue, denatures proteins.
    - **Enteroendocrine cells / G cells**: enteroendocrine cells numerous in fundus (scattered among parietal/chief cells), occasional in cardia/pylorus. **G cells** (in gastric pits of pyloric part) secrete **gastrin**, released when food enters stomach; stimulates parietal + chief cell secretory activity and promotes gastric smooth muscle mixing/churning.
    - **Chief cells**: found only in fundus, most abundant at gastric gland base; columnar; secrete **pepsinogen** (inactive proenzyme) -- acid converts it to **pepsin** (active proteolytic enzyme, optimal pH 1.5-2.0). Newborn stomachs (not adults) also produce **rennin** (coagulates milk proteins) and **gastric lipase** (digests milk fats).
- Stomach lining does not directly absorb nutrients, but some salts, water, and lipid-soluble substances (e.g., alcohol, aspirin, NSAIDs) are indirectly absorbed after damaging surface epithelium, entering bloodstream via lamina propria capillaries.

Regulation of Gastric Activity

- Direct control: (1) CNS via vagus nerve (parasympathetic) and celiac plexus branches (sympathetic) -- sight/thought of food triggers parasympathetic motor output to parietal/chief/mucous cells; (2) local/hormonal -- food arrival stimulates stretch receptors and mucosal chemoreceptors -> reflexive muscular contractions + G cell gastrin release; parietal and chief cells respond to gastrin (parietal cells especially sensitive, so acid production rises more than enzyme secretion).
- Sympathetic activation inhibits gastric activity. Small intestine hormones **secretin** and **cholecystokinin (CCK)** inhibit gastric secretion while stimulating pancreatic and liver secretion (a secondary/complementary effect).
- **Clinical: Gastritis** -- inflammation/irritation of stomach lining. **Acute gastritis**: sudden onset, often viral ("stomach flu") or food poisoning; fever, cramping, nausea, vomiting, diarrhea within 2-4 hrs of contaminated food. **Chronic gastritis**: often caused by Helicobacter pylori infection; untreated can lead to peptic ulcer.
- **Clinical: Peptic Ulcer Disease** -- sores of alimentary mucosa exposed to acidic secretions; usually stomach (gastric ulcers) or duodenum (duodenal ulcers). Causes: excess alcohol, smoking/chewing tobacco, aspirin/NSAIDs, caffeine, severe illness, radiation. Diagnosed/sometimes treated via endoscopy; H. pylori infections treated with antibiotics; acid-neutralizing/blocking meds help; avoid alcohol/tobacco/aspirin/NSAIDs. Untreated ulcers can erode through epithelium -> expose lamina propria -> pain/bleeding -> possible perforation -> gastric contents leak into peritoneum -> surgical emergency.

25.6 | The Small Intestine

- ~90% of nutrient absorption occurs in small intestine; most of the rest in proximal large intestine.
- Average length **6 m (20 ft)** (range 5.0-8.3 m [5-25 ft]); diameter ~**4 cm (1.6 in.)** at stomach junction, ~**2.5 cm (1 in.)** at large intestine junction.
- Found in all abdominal regions except left hypochondriac and epigastric regions. Held in place by mesenteries attached to dorsal body wall.
- **Circular folds (plicae circulares)**: permanent ring-shaped mucosal projections (unlike gastric rugae, don't disappear when filled); ~**800 folds** total (~2 per cm) along duodenum, jejunum, proximal half of ileum.
- Mucosa has **intestinal villi**; each epithelial cell has apical **microvilli**.
- Three anatomical subdivisions:
    - **Duodenum**: shortest/widest segment, ~**25 cm (10 in.)** long. Connected to pylorus via pyloric sphincter. Curves in a C around the pancreas. Proximal 2.5 cm (1 in.) is intraperitoneal; rest is secondarily retroperitoneal, located between vertebrae **L1-L4**. Functions as "mixing bowl" receiving chyme from stomach. No supporting mesentery (except the mobile proximal 2.5 cm).
    - **Jejunum**: begins at **duodenojejunal flexure**; re-enters peritoneal cavity, becomes intraperitoneal, supported by **mesentery proper**. ~**2.5 m (8 ft)** long. Bulk of chemical digestion and nutrient absorption occurs here. Thicker wall, larger diameter than ileum; typically in umbilical region; vasculature from mesentery is more straightforward/fewer branches than ileum's.

  - **Ileum**: third/last segment; intraperitoneal; longest segment, averaging **3.5 m (12 ft)**. Ends at **ileocecal valve** (sphincter controlling flow into cecum; protrudes into cecum). Typically lies in hypogastric region. Mesentery covering ileum thicker, contains more fat than jejunum's mesentery.
- Blood vessels/lymphatics/nerves of jejunum & ileum pass through mesentery proper -- branches of **superior mesenteric artery** and tributaries of **superior mesenteric vein**. Vagus nerve = parasympathetic innervation; superior mesenteric ganglion = sympathetic innervation.

Histology of the Small Intestine

- **Intestinal villi**: fingerlike mucosal projections into lumen, covered by simple columnar epithelium; apical surfaces have **microvilli**. Circular folds + villi + microvilli increase surface area by a factor of **>600**, to ~**2 million cm2**.
- **Goblet cells** between columnar epithelial cells secrete mucin onto intestinal surface.
- **Intestinal glands (crypts)**: at villus bases, extend into lamina propria; stem cells near gland base continually divide, producing new epithelial cells that migrate to villus tip within days, renewing surface and adding intracellular enzymes to chyme.
  - **Paneth cells**: at gland base; role in innate immunity; release defensins and lysozyme -- kill some bacteria, allow others to survive, establishing luminal flora.
  - Intestinal glands also contain enteroendocrine cells producing intestinal hormones.
- **Lamina propria**: loose areolar connective tissue in villus core; contains lymphatic cells, occasional lymphoid nodules, extensive capillary network (absorbs/carries nutrients to hepatic portal circulation). Each villus has a central **lacteal** (lymphatic capillary) that transports materials too large for blood capillaries -- e.g., **chylomicrons** (protein-lipid packages of absorbed fatty acids), which reach venous circulation via the thoracic duct.
- **Duodenum-specific histology**: submucosa contains **duodenal submucosal glands (Brunner's glands)** -- produce large quantities of mucus (protects epithelium from chyme acidity, contains bicarbonate to raise pH); most abundant in proximal duodenum, decreasing toward jejunum. Intestinal pH changes from 1-2 to 7-8 roughly halfway along the duodenum. **Duodenal ampulla**: muscular chamber within duodenal wall where bile duct (from liver/gallbladder) and pancreatic duct (from pancreas) unite; opens into duodenum at the **duodenal papilla**. **Hepatopancreatic sphincter**: encircles bile duct lumen and (generally) pancreatic duct/ampulla.
- **Jejunum & ileum histology**: circular folds/villi prominent in proximal jejunum (most nutrient absorption here); folds/villi diminish toward ileum end (parallels declining absorptive activity). Distal ileum lacks circular folds; lamina propria contains **20-30 masses of aggregated lymphoid nodules (Peyer's patches)**, most abundant in terminal ileum near large intestine -- protect small intestine from large-intestine bacteria.

Regulation of the Small Intestine

- Weak peristaltic contractions slowly move materials while absorption occurs; controlled by neural reflexes involving submucosal and myenteric plexuses. Parasympathetic stimulation increases reflex sensitivity, accelerates peristalsis/segmentation (mixing within a few cm of stimulus). Food entering stomach triggers coordinated intestinal movements moving chyme toward large intestine; ileocecal valve opens to allow passage into large intestine.
- Secretions = **intestinal juice**; triggered by local short reflexes and parasympathetic (vagal) stimulation; sympathetic stimulation inhibits secretion. Duodenal enteroendocrine cells produce **secretin** and **cholecystokinin (CCK)**, coordinating secretory activities of stomach, duodenum, liver, and pancreas.

25.7 | The Large Intestine

- Horseshoe-shaped; begins at ileum junction, ends at anus; lies inferior to stomach/liver, forms almost complete frame around small intestine. Average length ~**1.5 m (5 ft)**, width ~**7.5 cm (3 in.)**. Three parts: (1) cecum, (2) colon, (3) rectum.
- Functions: (1) absorbs water and electrolytes, compacts contents into feces, (2) absorbs vitamins produced by bacteria, (3) stores feces before defecation.
- Blood supply: branches of superior and inferior mesenteric arteries; venous drainage via superior and inferior mesenteric veins.

Cecum and Appendix

- **Cecum**: expanded pouch receiving material from ileum (ileum attaches medially, opens at **ileocecal valve**); collects/stores material, begins compaction. Intraperitoneal.
- **Appendix (vermiform appendix)**: slender, hollow, ~**9 cm (3.6 in.)** long (variable), attached to posteromedial cecum surface; suspended by **meso-appendix**. Primary function = lymphatic organ. Inflammation = **appendicitis**. Intraperitoneal.

The Colon -- subdivided into 4 regions: ascending, transverse, descending, sigmoid.

- **Haustra**: pouches in colon wall permitting distension/elongation; internal folds correspond to creases between haustra.
- **Teniae coli**: three separate longitudinal smooth muscle bands on outer colon surface (= the outer longitudinal muscular layer in other organs); their muscle tone creates the haustra.
- **Omental appendices (fatty appendices of colon)**: teardrop-shaped fat sacs in the serosa.
- **Ascending colon**: begins at cecum's superior border, ascends along right lateral/posterior wall to inferior liver surface, turns left at **right colic (hepatic) flexure**. Secondarily retroperitoneal (visceral peritoneum covers lateral/anterior surfaces only).
- **Transverse colon**: curves anteriorly from hepatic flexure, crosses abdomen right to left. Initial segment intraperitoneal (supported by transverse mesocolon, separated from anterior wall by greater omentum layers); becomes secondarily retroperitoneal as it passes inferior to stomach's greater curvature. **Gastrocolic ligament** attaches it to stomach's greater curvature. Turns at **left colic (splenic) flexure** near spleen -> becomes descending colon.
- **Descending colon**: secondarily retroperitoneal; proceeds inferiorly along left abdomen; curves at iliac fossa to become sigmoid colon.
- **Sigmoid colon**: S-shaped, ~**15 cm (6 in.)** long; lies posterior to urinary bladder; suspended by sigmoid mesocolon; empties into rectum.

Rectum and Anus

- **Rectum**: last **15 cm (6 in.)** of digestive tract; expandable, temporary feces storage. Movement of feces into rectum triggers urge to defecate.
- **Anal canal**: last portion of rectum; contains **anal columns** (small longitudinal folds) whose distal margins join via transverse folds marking the boundary between columnar epithelium (proximal rectum) and stratified squamous epithelium (like oral cavity). **Anus** = exit of anal canal; epidermis here is keratinized, identical to skin surface.
- **Internal anal sphincter**: circular smooth muscle, involuntary. **External anal sphincter**: skeletal muscle ring around distal anal canal, voluntary control.
- Lamina propria/submucosa of anal canal contain a venous network; elevated venous pressure (straining during defecation, pregnancy) -> distended veins -> **hemorrhoids**.

Histology of the Large Intestine -- distinguishing features vs. small intestine:

- Wall much thinner than small intestine, despite colon diameter being ~3x that of small intestine.
- Lacks villi.
- Goblet cells more abundant in mucosal epithelium.
- Intestinal glands are deeper, with more goblet cells; secretion via local reflexes (local neural plexuses) produces large amounts of mucus lubricating compacted waste.
- Large lymphoid nodules scattered throughout lamina propria, extending into submucosa.
- Muscular layer's longitudinal component reduced to the three teniae coli bands; mixing/propulsive contractions otherwise resemble small intestine's.

Regulation of the Large Intestine

- Movement from cecum to transverse colon is very slow -> allows water absorption, converting material to sludgy paste.

- Peristaltic waves move material along colon length. **Haustral churning**: segmentation-like movements mixing contents of adjacent haustra.
- **Mass movements**: powerful peristaltic contractions triggered when stomach/duodenum are distended (signaled via intestinal neural plexuses); move material from transverse colon through rest of large intestine, forcing feces into rectum.
- **Defecation reflex**: rectal distension (from mass movement forcing feces into rectum) stimulates stretch receptors -> relaxes internal anal sphincter, feces move into anal canal; voluntary relaxation of external anal sphincter permits defecation.
- **Clinical case (Tayvian) -- C. difficile colitis**: prolonged antibiotic use kills normal "good" colonic bacteria (microbiome that aids digestion and vitamin generation), allowing Clostridium difficile to proliferate and produce toxins causing colitis (colon inflammation). Symptoms: watery/bloody diarrhea (up to 10x/day), no appetite, crampy abdominal pain, fever, weight loss. Treatment: **fecal microbiota transplant** from a healthy donor (e.g., relative) -- donated feces (containing healthy bacteria) infused via NG tube into duodenum; bacteria recolonize the large intestine, crowd out C. diff, restore normal digestion/vitamin production.
    - **Bristol Stool Chart** (7 types): Type 1 = separate hard lumps (very constipated); Type 2 = lumpy sausage-shaped (slightly constipated); Type 3 = sausage shape with surface cracks (normal); Type 4 = smooth soft sausage/snake (normal); Type 5 = soft blobs with clear-cut edges (lacking fiber); Type 6 = mushy, ragged edges (inflammation); Type 7 = liquid, no solid pieces (inflammation).

25.8 | Accessory Digestive Organs

- Liver, gallbladder, pancreas produce/store enzymes and buffers essential for digestion. Liver and pancreas also have exocrine functions.

The Liver

- Functions: (1) regulates metabolism, (2) regulates blood composition (removes old/damaged RBCs, synthesizes plasma proteins), (3) produces/secretes bile.
- Largest visceral organ; firm, reddish-brown; weighs ~**1.5 kg (3.3 lb)**. Most mass in right hypochondriac and epigastric regions, may extend into left hypochondriac and umbilical regions.
- **Metabolic regulation**: regulates circulating carbohydrate, lipid, amino acid levels. Blood from all absorptive GI surfaces enters via **hepatic portal system** -> liver extracts nutrients/toxins before reaching systemic circulation (via hepatic veins). **Hepatocytes** monitor/adjust metabolite levels -- store excess nutrients, mobilize reserves or synthesize to correct deficiencies. Removes/inactivates/stores/excretes circulating toxins and metabolic wastes. Absorbs and stores fat-soluble vitamins (A, D, K, E).
- **Hematological regulation**: largest blood reservoir in the body, receiving ~**25% of cardiac output**. Phagocytic cells remove old/damaged RBCs, cellular debris, pathogens as blood passes through. Liver cells synthesize/secrete plasma proteins (osmotic concentration, nutrient transport, clotting and complement systems).
- **Bile production**: synthesized by hepatocytes, stored in gallbladder, excreted into duodenal lumen. Bile = mostly water + ions + **bilirubin** (hemoglobin-derived pigment) + bile salts (lipids). Water/ions dilute and buffer chyme acids; bile salts enable enzymatic breakdown of lipids into absorbable fatty acids.
- Liver has >200 known functions (partial list per textbook table includes): synthesize somatomedins; synthesize/secrete bile; store glycogen/lipid reserves; maintain blood glucose/amino acid/fatty acid levels; synthesize/convert nutrients (transaminate amino acids, convert carbs to lipids); synthesize/release cholesterol bound to transport proteins; inactivate toxins; store iron reserves; store fat-soluble vitamins; synthesize plasma proteins; synthesize clotting factors; synthesize inactive angiotensinogen; phagocytose damaged RBCs (via stellate macrophages); store blood (major venous reserve contributor); absorb/break down circulating hormones (incl. insulin, epinephrine) and immunoglobulins; absorb/inactivate lipid-soluble drugs.
- Severe liver damage is life-threatening. Liver cells have limited regenerative ability after injury, but full function requires normal vascular pattern regeneration too.
- Intraperitoneal; deep to visceral peritoneum is a tough fibrous capsule. **Falciform ligament** (ventral mesentery) marks division between left and right lobes on anterior surface. **Round ligament (ligamentum teres)**:

thickening of falciform ligament's inferior margin, marking the path of the degenerated fetal umbilical vein. **Coronary ligament**: suspends liver from inferior diaphragm surface.

- Surfaces: anterior/superior/posterior = **diaphragmatic surfaces**; inferior = **visceral surface** (bears impressions from stomach, small intestine, right kidney, large intestine).
- Classical 4-lobe description: **left lobe**, **right lobe**, **caudate lobe** (small, marked off by IVC impression, divides right lobe region), **quadrate lobe** (inferior to caudate, sandwiched between left lobe and gallbladder).
- Newer terminology subdivides lobes into segments based on subdivisions of hepatic artery, portal vein, and hepatic ducts (developed to meet surgical needs).
- **Blood supply**: two vessels deliver blood -- **hepatic artery proper** (~1/3 of hepatic blood flow, oxygen-rich) and **hepatic portal vein** (~2/3 of flow, nutrient/chemical-rich from intestines). Blood returns via **hepatic veins** -> inferior vena cava.

Histology of the Liver

- Each lobe divided by connective tissue into ~**100,000 liver lobules** -- the basic functional units.
- **Liver lobule**: hepatocytes form irregular plates like spokes of a wheel. Up to age 7, plates are up to two cells thick; after age 7, plates are one cell thick. Hepatocyte apical/basal surfaces covered with short microvilli.
- Blood from hepatic artery proper + hepatic portal system drains into fenestrated capillaries called **hepatic sinusoids** surrounding hepatocyte plates. Fenestrated (large-opening) walls allow substances to pass into spaces around hepatocytes.
- **Stellate macrophages (Kupffer cells)**: numerous phagocytic cells in sinusoidal lining, part of monocyte-macrophage system; engulf pathogens, cell debris, damaged blood cells; also store iron, some lipids, heavy metals (tin, mercury) absorbed by the digestive tract.
- Lobule is hexagonal in cross-section; **6 portal triads**, one at each corner. Each **portal triad** = (1) interlobular vein, (2) interlobular artery, (3) interlobular bile duct.
- Blood flows from triad branches into sinusoids of adjacent lobules -> hepatocytes absorb plasma solutes and secrete materials (e.g., plasma proteins) -> blood leaves sinusoids into the **central vein** of the lobule -> central veins merge into **hepatic veins** -> inferior vena cava.
- **Bile secretion/transport**: hepatocytes form bile, secreted into narrow **bile canaliculi** between adjacent liver cells -> extend to fine **bile ductules** -> carry bile to an **interlobular bile duct** in the nearest portal triad -> right and left **hepatic ducts** collect bile from all lobe ducts -> unite to form the **common hepatic duct** leaving the liver -> bile flows either into the **bile duct** (empties at duodenal ampulla) or into the **cystic duct** (leads to gallbladder).

The Gallbladder

- Hollow, pear-shaped, muscular sac; stores and concentrates bile before release into small intestine. Located in a fossa on the posterior surface of the liver's right lobe. Intraperitoneal.
- Three regions: **fundus**, **body**, **neck**. **Cystic duct** exits gallbladder, unites with common hepatic duct to form the **bile duct**.
- At the duodenum, the **sphincter of ampulla** surrounds the lumen of the bile duct and duodenal ampulla; opens at the **duodenal papilla**. Contraction seals the passage, preventing bile entry into small intestine.
- Two major functions: storing and modifying (concentrating) bile. When sphincter of ampulla is closed, bile flows into cystic duct -> gallbladder for storage. Capacity when filled: **40-70 mL** of bile. While stored, water is absorbed from bile, concentrating bile salts and other components.
- **Bile ejection**: stimulated by hormone **cholecystokinin (CCK)**, released into blood at the duodenum when chyme contains large amounts of lipids/partially digested proteins. CCK relaxes the sphincter of ampulla and contracts the gallbladder.
- **Histology**: wall = mucosa, lamina propria, muscular layer, serosa only -- lacks a muscularis mucosae and submucosa. Muscular layer = two interlacing smooth muscle layers (inner mostly longitudinal, outer mostly circular).

The Pancreas

- Lies posterior to stomach, extends laterally from duodenum toward spleen. Elongated, pinkish-gray; ~**15 cm (6 in.)** long, ~**80 g (3 oz)**. Three subdivisions: **head, body, tail** (tail is short/rounded, near spleen). Secondarily retroperitoneal, firmly bound to posterior abdominal wall.
- Nodular/lumpy surface texture; thin transparent connective tissue capsule; pancreatic lobules, blood vessels, excretory ducts visible through capsule and overlying peritoneum.
- **Mixed gland**: exocrine (digestive enzymes + buffers) and endocrine functions (endocrine function detailed in Ch 19, p. 518-520 -- see below).
- **Pancreatic duct** (main duct): delivers exocrine secretions to the duodenal ampulla. **Accessory pancreatic duct (duct of Santorini)**: sometimes present, branches off before main duct leaves pancreas; when present, typically empties at a separate duodenal papilla outside the ampulla.
- Blood supply: splenic, superior mesenteric, and common hepatic arteries (via pancreatic arteries and superior/inferior pancreaticoduodenal arteries); venous drainage via splenic vein and branches.

Histological Organization of the Pancreas

- Connective tissue septa divide pancreas into lobules; blood vessels and duct tributaries run within septa. Compound tubulo-acinar gland.
- **Pancreatic acini**: secretory units, simple cuboidal epithelium; beginning of the duct system, draining into progressively larger ducts -> main pancreatic duct.
- Acini produce **pancreatic juice** (water, ions, digestive enzymes) released into duodenum to break down materials for absorption. Pancreatic ducts also secrete buffers (primarily **sodium bicarbonate**) to neutralize chyme acid and stabilize intestinal pH.
- **Pancreatic islets** scattered among acini -- account for ~**1%** of pancreatic cell population; responsible for endocrine function (see Ch 19 notes).
- **Pancreatic enzymes** by target:
    - **Lipases**: digest lipids.
    - **Carbohydrases** (e.g., pancreatic amylase): digest sugars/starches.
    - **Nucleases**: break down nucleic acids.
    - **Proteolytic enzymes**: break proteins apart -- includes **proteinases** (break large protein complexes) and **peptidases** (break small peptide chains into amino acids).
- Regulation: duodenal hormones trigger pancreatic juice secretion. **Secretin** (released when acid chyme arrives) triggers watery, buffer-rich pancreatic juice. **Cholecystokinin (CCK)** stimulates production/secretion of pancreatic enzymes.

25.9 | Aging and the Digestive System

- Digestion/absorption remain essentially normal in elderly, but changes parallel age-related changes in other systems:
    - **Cumulative damage** becomes apparent -- e.g., tooth loss from decay/gingivitis; toxins (alcohol, heavy metals, chemicals) processed/stored by liver can cause chronic damage -> cirrhosis or other liver disease.
    - **Epithelial stem cell division rate declines** -- less efficient tissue repair; digestive epithelium more susceptible to abrasion/acid/enzyme damage; peptic ulcers more likely; stratified epithelium (mouth, esophagus, anus) becomes thinner/more fragile.
    - **Smooth muscle tone decreases**, motility decreases, peristaltic contractions weaken -- slows chyme movement, promotes constipation. Sagging haustral walls -> **diverticulitis** symptoms. Straining to defecate stresses vessel walls -> **hemorrhoids**. Weakened cardiac sphincter -> esophageal reflux/frequent heartburn.
    - **Cancer rates increase**, especially in organs with actively dividing stem cell populations -- colon and stomach cancer rates rise; oral/pharyngeal cancers particularly common in elderly smokers.
    - Changes in other systems affect digestion -- e.g., reduced bone mass/calcium -> erosion of tooth sockets -> tooth loss; declining olfactory/gustatory sensitivity -> dietary changes affecting whole body.

Related Clinical Terms (Ch 25)

- **Crohn disease**: incurable chronic inflammatory bowel disease, can affect any part of the digestive tract (mouth to anus); strictures, fistulas, fissures common.
- **Irritable bowel syndrome (IBS)**: common large-intestine disorder with cramping, abdominal pain, bloating, gas, diarrhea, constipation.
- **Esophageal varices**: swollen, fragile esophageal veins resulting from portal hypertension.
- **Pancreatic cancer**: malignancy often asymptomatic early, leading to late detection; survival rate only ~4%.
- **Gastrectomy**: surgical removal of the stomach, generally for advanced stomach cancer.
- **Polyps**: small growths with a stalk protruding from a mucous membrane, usually benign.
- **Pyloric stenosis**: uncommon condition where lower stomach muscle enlarges, preventing food from entering the small intestine.

Ch 26 — The Urinary System

Introduction

- Urinary system works with digestive, cardiovascular, and respiratory systems to prevent "pollution" problems inside the body; removes most metabolic wastes generated by body cells.
- Essential functions: (1) regulating blood plasma concentrations of sodium, potassium, chloride, calcium, and other ions by controlling quantities excreted in urine; (2) regulating blood volume and blood pressure by (a) adjusting volume of water lost in urine, (b) releasing erythropoietin, (c) releasing renin; (3) working with respiratory system to regulate blood pH; (4) conserving valuable nutrients by preventing their excretion in urine; (5) eliminating metabolic wastes (especially urea, uric acid, toxic substances); (6) synthesizing calcitriol (vitamin D hormone derivative that stimulates intestinal calcium absorption); (7) helping the liver detoxify poisons; (8) during starvation, deaminating amino acids for use by other tissues.
- Urinary system = kidneys, ureters, urinary bladder, urethra. Kidneys perform excretory functions, producing urine (fluid waste with water, ions, small soluble compounds). Urinary tract = paired ureters, urinary bladder, urethra. Urine temporarily stored in bladder; urination (micturition) = muscular bladder contracts to force urine through urethra.

26.1 | The Kidneys

- Kidneys lie lateral to vertebral column, between last thoracic vertebra and third lumbar vertebra (T12-L3) on each side. Right kidney sits slightly lower than left due to the liver.
- Anterior surface of right kidney covered by liver, right colic (hepatic) flexure, duodenum. Anterior surface of left kidney covered by spleen, stomach, pancreas, jejunum, left colic (splenic) flexure.
- Adrenal (suprarenal) gland sits on superior surface of each kidney. Kidneys, adrenal glands, and ureters are all retroperitoneal (lie against posterior body wall muscles).
- Position maintained by: (1) overlying parietal peritoneum contacts with adjacent visceral organs, (2) supporting connective tissues. Three concentric connective tissue layers protect/stabilize each kidney:

1. **Fibrous capsule**: covers outer surface of entire organ; collagen fiber layer maintains shape, provides physical protection.

2. **Perinephric fat (perirenal fat capsule)**: adipose layer surrounding fibrous capsule.

3. **Renal fascia**: dense outer connective tissue layer; collagen fibers extend from inner fibrous capsule through perinephric fat to renal fascia; anchors kidney to surrounding structures, attaches to deep fascia of posterior body wall muscles. **Pararenal fat body**: another adipose layer separating posterior/lateral renal fascia from body wall. Anteriorly, renal fascia attaches to peritoneum and to the anterior renal fascia of the opposite side.

- This arrangement cushions kidneys against day-to-day jolts. If suspensory fibers stretch or adipose padding is reduced, kidneys become more vulnerable to traumatic injury (nephroptosis / "floating kidney").

Superficial Anatomy of the Kidney

- Each kidney is kidney-bean shaped, brownish-red; typical adult kidney ~**10 cm (4 in.)** long, **5.5 cm (2.2 in.)** wide, **3 cm (1.2 in.)** thick; averages ~**150 g (5.30 oz)**.
- **Hilum**: medial indentation; renal blood vessels, lymphatic vessels, nerves, and the ureter pass through it and branch within the **renal sinus** (internal cavity lined by the inner layer of the fibrous capsule, which folds inward at the hilum). Outer (thick) layer of the capsule extends across the hilum and stabilizes these structures' position.

Sectional Anatomy of the Kidney

- Deep to the renal capsule: **renal cortex** — granular, reddish-brown.
- Deep to cortex: **renal medulla** — consists of **6–18 renal pyramids** (triangular structures). Base of each pyramid faces the cortex; tip (**renal papilla**) projects into the renal sinus. Each pyramid has fine grooves converging at the papilla.
- **Renal columns**: bands of cortical tissue extending into the medulla, separating adjacent pyramids.
- **Kidney lobe** = one renal pyramid + overlying renal cortex + adjacent renal column tissue. Urine is produced within kidney lobes.
- Urine drainage pathway: renal papilla ducts → **minor calyx** (cup-shaped drain) → 4–5 minor calyces merge → **major calyx** → major calyces combine → **renal pelvis** (large funnel-shaped chamber) → connects to the **ureter**.
- Two nephron types: **cortical nephrons** (in renal cortex) and **juxtamedullary nephrons** (closer to renal medulla). Each kidney has roughly **1.25 million nephrons**, combined length ~**145 km (90 miles)**.

The Blood Supply to the Kidneys

- **20–25% of total cardiac output** (~1200 mL/min) flows through the kidneys.
- Pathway: **renal artery** (branches from lateral abdominal aorta near superior mesenteric artery) → enters renal sinus → **segmental arteries** → **interlobar arteries** (radiate through renal columns between pyramids) → **arcuate arteries** → **cortical radiate (interlobular) arteries** (supply kidney lobe portions) → numerous **afferent arterioles** (supply individual nephrons).
- Venous return (mirror image, no segmental veins): nephrons → venules/small veins → **cortical radiate (interlobular) veins** → **arcuate veins** → **interlobar veins** → merge to form the **renal vein**.

Innervation of the Kidneys

- Urine production regulated by altering nephron filtration rates via **autoregulation** (local blood flow regulation) — reflexive changes in arteriole diameter altering blood flow/filtration rates.
- **Renal nerves** innervate kidneys and ureters — mostly sympathetic postganglionic fibers from celiac and inferior mesenteric ganglia; enter kidney at hilum, follow renal artery branches to individual nephrons.
- Sympathetic innervation: (1) adjusts urine formation rate by changing nephron blood flow, (2) influences urine composition by stimulating renin release.

Histology of the Kidney — The Nephron and Collecting System

- **Nephron** = basic structural/functional unit of the kidney; consists of **renal corpuscle** + **renal tubule**.
- **Renal corpuscle**: spherical structure = **glomerular capsule** (cup-shaped chamber) + **glomerulus** (capillary network).
- **Renal tubule**: long tubular passageway beginning at renal corpuscle, three subdivisions: (1) **proximal convoluted tubule (PCT)**, (2) **nephron loop (loop of Henle)**, (3) **distal convoluted tubule (DCT)**.
- Each nephron empties into the **collecting system**: **connecting tubule** (carries filtrate from DCT to a nearby **collecting duct**) → collecting duct leaves cortex, descends into medulla → carries filtrate to a **papillary duct** → drains into renal pelvis.
- **Cortical nephrons**: ~**85%** of all nephrons; located almost entirely in superficial cortex; short nephron loop; efferent arteriole → **peritubular capillaries** surrounding the renal tubule → drain to interlobular veins. Because more numerous, cortical nephrons perform most reabsorptive/secretory functions.

- **Juxtamedullary nephrons**: remaining ~**15%**; located closer to medulla; longer nephron loops extending deep into renal pyramids; create conditions necessary for producing concentrated urine.
- Filtration alone is passive and size-based — cannot distinguish wastes from useful small molecules (water, ions, glucose, fatty acids, amino acids). Distal nephron segments are responsible for: (1) reabsorbing useful metabolic substrates from filtrate, (2) reabsorbing >80% of filtrate water, (3) secreting missed waste products into filtrate.

The Renal Corpuscle

- Averages **150–250 µm** diameter. Includes glomerulus capillaries + **glomerular capsule (Bowman's capsule)**. Glomerulus projects into the capsule (like heart into pericardial cavity).
- **Capsular outer layer (parietal layer)**: simple squamous epithelium, outer capsule wall.
- **Visceral layer**: covers glomerular capillaries; composed of **podocytes —** large cells with complex processes wrapping around glomerular capillaries.
- **Capsular space**: separates parietal and visceral epithelia.
- **Vascular pole**: where parietal/visceral epithelia connect and glomerular capillaries connect to afferent/efferent arterioles. Blood arrives via **afferent arteriole**, exits via smaller-diameter **efferent arteriole**.
- **Filtration**: blood pressure forces fluid/dissolved solutes out of glomerular capillaries into capsular space → filtrate resembles plasma minus blood proteins.
- **Five filtration barriers**:

1. **Endothelial surface layer**: luminal surface of glomerular capillary endothelial cells has a thick carbohydrate-rich glycocalyx meshwork limiting large plasma protein filtration.

2. **Capillary endothelium**: fenestrated capillaries with pores **60–100 nm (0.06–0.1 µm)** diameter — small enough to block blood cells, but allow diffusion of solutes up to size of... (small proteins).

3. **Basement membrane**: several times thicker than typical capillary basement membrane; blocks larger plasma proteins but permits small plasma proteins, amino acids, glucose, ions. Unlike other cardiovascular basement membranes, this one encircles two or more capillaries **— mesangial cells** located between endothelial cells of adjacent capillaries provide physical support and regulate glomerular blood flow/filtration by engulfing organic materials and regulating capillary diameters.

4. **Glomerular epithelium**: podocytes have long processes wrapping around the outer basement membrane surface; secondary processes ("feet") separated by narrow **filtration slits** — filtrate consists of water with dissolved ions, small molecules, few if any plasma proteins.

5. **Subpodocyte space**: occupies ~**60%** of the glomerulus filtration space; narrow space between podocyte secondary processes and podocyte soma; assists filtration slits.

- Filtrate also contains glucose, free fatty acids, amino acids, vitamins — reabsorbed by the PCT.

The Proximal Convoluted Tubule (PCT)

- First part of renal tubule; begins at the tubular pole of the renal corpuscle (opposite vascular pole). Lined by simple cuboidal epithelium with apical microvilli (increase surface area for reabsorption).
- Primary function = **reabsorption**. Actively reabsorbs organic nutrients, ions, plasma proteins (if any). Reabsorbs **60%** of sodium ions, chloride ions, and water; also actively reabsorbs potassium, calcium, magnesium, bicarbonate, phosphate, sulfate ions. Osmotic forces pull water across the PCT wall into peritubular (interstitial) fluid as solutes are reabsorbed.

The Nephron Loop

- Proximal straight tubule ends as renal tubule enters medulla — bend marks the nephron loop start. Divided into **descending limb** and **ascending limb**; fluid in ascending limb flows toward renal cortex. Both limbs lined by simple squamous epithelium, found in the deeper medulla.
- Ascending limb has active transport mechanisms pumping sodium and chloride ions out of tubular fluid → medullary interstitial fluid has unusually high solute concentration. Limbs contain thick and thin segments (referring to epithelium height, not lumen diameter).
- Solute concentration expressed in **milliosmoles (mOsm)**. Near the loop base (deepest medulla), interstitial fluid solute concentration is **four times that of plasma** (**1200 mOsm vs. 300 mOsm**).

- Descending thin limb and ascending thick limb: freely permeable to water but impermeable to ions/solutes. High osmotic concentration surrounding the loop causes osmotic flow of water out of the nephron. **Vasa recta** (slender capillaries) absorb this water, returning it to general circulation.
- Net effect: nephron loop reabsorbs an additional **25%** of tubular fluid water plus even more sodium/chloride. Combined PCT + nephron loop reabsorption reclaims all organic nutrients, **85%** of water, and >**90%** of sodium and chloride. Remaining water, ions, and all organic wastes filtered at the glomerulus enter the DCT.

The Distal Convoluted Tubule (DCT)

- Ascends out of medulla into cortex; passes between afferent and efferent arterioles at the vascular pole. Differs from PCT: (1) smaller diameter, (2) epithelial cells lack microvilli, (3) cell boundaries easily seen. PCT = reabsorption; DCT = secretion.
- DCT functions: (1) active secretion of ions, acids, other materials; (2) reabsorption of sodium and calcium ions from tubular fluid; (3) reabsorption of water (concentrates tubular fluid). Sodium transport activity of DCT controlled by circulating **aldosterone** (secreted by adrenal cortex).

The Juxtaglomerular Complex

- Structure that helps regulate blood pressure and filtrate formation; secretes **renin** and **erythropoietin**, which elevate blood volume, hemoglobin levels, and blood pressure, restoring normal filtrate production rates. Composed of three specialized cell types:

1. **Macula densa**: (region of the DCT near the vascular pole) thought to monitor Na+ concentration in tubular fluid and regulate both glomerular filtration rate and renin release from juxtaglomerular cells.

2. **Juxtaglomerular cells**: modified smooth muscle cells in the afferent arteriole wall that secrete renin.

3. **Extraglomerular mesangial cells**: located in the triangular space between afferent and efferent glomerular arterioles; provide feedback control between the macula densa and juxtaglomerular cells.

The Collecting System

- DCT (last nephron segment) opens into the collecting system: **connecting tubules** → **collecting ducts** → **papillary ducts**. Every nephron connects to a connecting tubule; several connecting tubules connect to one collecting duct; several collecting ducts converge into a larger papillary duct emptying into a minor calyx.
- Epithelium: begins as simple cuboidal in connecting tubules, changes to columnar in collecting/papillary ducts.
- Transports tubular fluid from nephron to renal pelvis, making final osmotic concentration/volume adjustments. **Antidiuretic hormone (ADH)** controls collecting system permeability. Because collecting ducts pass through the medulla (where the nephron loop established very high solute concentrations), low permeability → dilute urine (most fluid flows to renal pelvis); high permeability → promotes osmotic water flow out of the duct into the medulla → small amount of highly concentrated urine. Higher circulating ADH → greater water reabsorption → more concentrated urine.

26.2 | Structures for Urine Transport, Storage, and Elimination

- Filtrate modification/urine production ends when fluid enters the minor calyx. Remaining urinary system parts (ureters, bladder, urethra) transport, store, eliminate urine.
- **Transitional epithelium** lines the minor and major calyces, renal pelvis, ureters, urinary bladder, and proximal urethra — tolerates distension/relaxation cycles without damage.

The Ureters

- Pair of retroperitoneal muscular tubes extending from kidneys to urinary bladder. Each ureter ~**30 cm (12 in.)** long; begins as a funnel-shaped continuation of the renal pelvis. Pass inferiorly/medially over the **psoas major**. Anatomical location differs between sexes due to reproductive organ variation.
- Ureters penetrate the posterior bladder wall obliquely; **ureteral orifices are slit-like (not rounded)** — prevents urine backflow toward kidneys when the bladder contracts.
- **Histology**: three wall layers: (1) inner mucosa (transitional epithelium), (2) middle muscular layer (inner longitudinal + outer circular smooth muscle), (3) outer adventitia (continuous with kidney's fibrous capsule and abdominal wall parietal peritoneum).

- Peristaltic contractions of the muscular wall occur roughly every **30 seconds**, triggered by stretch receptor stimulation in the ureteral wall; travel from renal pelvis to bladder, "milking" urine through the ureter.
- **Clinical: Kidney stones/renal colic (Clinical Case — Jack)**: a kidney stone = small, hard, sharp-edged mineral deposit (usually calcium and phosphate) that normally stays dissolved in urine; forms when urine becomes overly concentrated (e.g., from dehydration/sweating). As the stone moves into the ureter, it obstructs urine flow, stretches the collecting system behind it (flank/lumbar pain), abrades the ureteral epithelium (causing **hematuria —** bloody urine), and triggers painful peristaltic contractions trying to push it along. Pain radiates as the stone passes over the psoas major, then to the groin/testicle/urethral tip as it nears the bladder. Treatment/prevention: increased fluid intake ("drink, drink, drink").
- **Clinical: Dialysis** — performs some normal kidney functions; required once **90% of normal kidney function** is lost (end-stage kidney disease). Removes wastes, salt, extra water; balances electrolytes/minerals/pH — but cannot produce renin, erythropoietin, or calcitriol. Hemodialysis: performed at a dialysis center ~3x/week for a few hours; requires venous access (surgical forearm shunt) connected to a machine that cleanses blood. **Peritoneal dialysis**: uses the peritoneal cavity as the dialysis membrane via a permanent intraperitoneal catheter; dialysate (~2 quarts) infused, dwells 4–5 hours, then drained/replaced (continuous ambulatory peritoneal dialysis, performed 4–5x/day by the patient) or cycled overnight by machine (automated peritoneal dialysis).
- Nearly half a million Americans are on dialysis; ~200,000 have a functioning transplanted kidney; >100,000 await transplant, but fewer than 20,000 transplants performed/year.
- **Acute renal failure**: sudden kidney function loss, often from multiple trauma, myocardial infarction, or reduced renal blood supply; often reversible with temporary dialysis during recovery.
- **Chronic renal failure**: gradual loss of kidney function, may not be apparent until significantly impaired. Diabetes and hypertension are leading causes; others include genetic diseases, infections, kidney stones, autoimmune diseases, certain drugs (legal/illegal, including some antibiotics).
- Diet management for kidney failure: strictly limit fluids, protein, and minerals (sodium, potassium, phosphorus).
- **Kidney transplant** = best treatment for renal failure, from living or deceased (within 48 hrs of death) donor. Recipient's nonfunctioning kidney(s) may be removed (especially if infected). Transplanted kidney/ureter usually placed extraperitoneally in the pelvic cavity (iliac fossa); ureter connected to recipient's bladder. 2-year survival rate >90% — better than dialysis-only survival rate.

The Urinary Bladder

- Hollow, muscular organ for temporary urine storage. In males, bladder base lies between rectum and pubic symphysis; in females, base sits inferior to uterus, anterior to vagina. Full bladder can hold ~**1 liter** of urine.
- Peritoneum covers superior surfaces of an empty bladder; as bladder fills, it displaces parietal peritoneum from the anterior abdominal wall and becomes intraperitoneal.
- Peritoneal folds stabilizing bladder position: **median umbilical ligament** (anterior/superior bladder border to umbilicus) and **lateral umbilical ligaments** (bladder sides to umbilicus) — fibrous remnants of the two umbilical arteries that supplied the placenta during development. Tough connective tissue bands also anchor posterior, inferior, anterior bladder surfaces to pelvic bones.
- Mucosa lining has folds (**rugae**) that disappear as bladder stretches/fills.
- **Trigone**: smooth, very thick mucosa region; acts as a funnel channeling urine into urethra when bladder contracts. Urethral entrance = most inferior point of bladder. **Neck of the urinary bladder** = region surrounding urethral opening, contains the muscular **internal urethral sphincter —** involuntary control via pelvic nerve branches.
- **Histology**: mucosa (transitional epithelium with prominent rugae) → submucosa (connective tissue) → muscular layer (**3 layers**: inner longitudinal, middle circular, outer longitudinal smooth muscle — together forming the **detrusor**) → adventitia (outermost layer). Detrusor contraction compresses the bladder and expels urine into the urethra.
- **Clinical: Transitional cell carcinoma** — most common bladder cancer type; begins in innermost transitional epithelium layer. Smoking = most important risk factor (toxins filtered by lungs/kidneys may still concentrate in

stored urine). Hallmark sign = **hematuria** (bloody urine). Diagnosed/treated via **cystoscopy** (thin lighted endoscope through urethra into bladder).

The Urethra

- Extends from bladder neck to exterior. Differs in length/function by sex.
- **Female urethra**: very short, **3–5 cm (1–2 in.)**, extends from bladder to vestibule; **external urethral orifice** near the anterior vaginal wall.
- **Male urethra**: extends from bladder neck to penis tip, **18–20 cm (7–8 in.)**; three portions: (1) **prostatic urethra** (through center of prostate), (2) **membranous urethra** (intermediate part; short segment penetrating pelvic cavity's muscular floor), (3) **spongy (penile) urethra** (from membranous urethra to external urethral orifice at penis tip).
- Both sexes: **external urethral sphincter** — skeletal muscle band acting as a valve; controlled by branches of the hypogastric plexus (both internal and external sphincters); only the external sphincter is under **voluntary control**, via the perineal branch of the pudendal nerve. Sphincter has resting muscle tone, must be voluntarily relaxed for urination. Autonomic innervation of external sphincter matters mainly when voluntary control is lacking (infants, adults with spinal cord injury).
- **Histology**: Female — transitional epithelium near bladder neck, stratified squamous epithelium for the rest; lamina propria has extensive venous network, surrounded by concentric smooth muscle layers. Male — varies along length: transitional near bladder → pseudostratified/stratified columnar → stratified squamous near external orifice; lamina propria thick/elastic, mucosa folded into longitudinal creases; mucus-secreting cells present, with epithelial mucous glands forming tubules into the lamina propria (males); connective tissue anchors urethra to surrounding structures.

Urinary Reflexes: Urine Storage and Urine Voiding

- Urine reaches bladder via ureteral peristaltic contractions. Urge to urinate generally appears when bladder contains ~**200 mL** of urine. Two spinal reflexes control micturition, working together: **urine storage reflex** and **urine voiding reflex**.
- **Urine storage reflex**: occurs via spinal reflexes and the **pontine storage center**. Low-frequency afferent impulses from bladder stretch receptors: (1) increase sympathetic activity (inhibits detrusor contraction, stimulates internal urethral sphincter contraction), (2) stimulate external urethral sphincter contraction. Pontine storage center also stimulates somatic motor neurons to the external sphincter, causing contraction. Together these promote continence (urine storage).
- **Urine voiding reflex**: occurs via spinal reflexes and the **pontine micturition center**. High-frequency afferent impulses from bladder stretch receptors stimulate interneurons relaying to the pontine micturition center, which initiates sacral spinal reflexes that: (1) increase parasympathetic activity (detrusor contracts, internal urethral sphincter relaxes), (2) decrease sympathetic activity (internal sphincter relaxes), (3) decrease somatic motor activity (external sphincter relaxes) → voiding occurs.
- At the end of typical urination, <**10 mL** of urine remains in the bladder.
- **Clinical: Urinary tract infections (UTIs)** — caused by bacteria, fungi, or viruses colonizing the urinary tract; *Escherichia coli* most often involved. Women more susceptible due to proximity of external urethral orifice to anus; sexual intercourse may push bacteria into the (relatively short) female urethra toward the bladder. Symptoms: **dysuria** (painful urination), fever, foul/strong-smelling urine, cloudy/bloody urine, frequent urination, persistent urge to urinate; in elderly, confusion is often the first symptom. Detected via bacteria/blood cells in urine. Inflamed urethral wall → impaired water/sodium reabsorption → more frequent urination, increased fluid requirements. Inflammation of urethra = **urethritis**; inflammation of bladder lining = **cystitis —** many infections affect both. Usually respond to antibiotics; mainstay therapy = dilute urine (drink fluids) + frequent urination (empty bladder often). Untreated: bacteria may migrate along ureters to renal pelvis → **pyelitis** (renal pelvis inflammation); if bacteria invade renal cortex/medulla too → **pyelonephritis** (high fever, intense unilateral pain, vomiting, diarrhea, blood cells/pus in urine).

26.3 | Aging and the Urinary System

- Aging is associated with increased kidney problems. Four typical age-related changes:

1. **Decline in functional nephron number**: between ages 25 and 85, total nephron count drops by **30–40%**.

2. **Reduced glomerular filtration**: from decreased glomeruli numbers, cumulative filtration apparatus damage in remaining glomeruli, and reduced renal blood flow.

3. **Reduced sensitivity to ADH**: distal nephron portions and entire collecting system become less responsive to ADH with age → reduced water/sodium reabsorption → more frequent urination, increased daily fluid requirements.

4. **Problems with urinary reflexes**: (a) external sphincter loses muscle tone, becomes less effective at voluntarily retaining urine → incontinence (often slow urine leakage); (b) ability to control urination can be lost due to stroke, Alzheimer's disease, or other CNS problems affecting cerebral cortex/hypothalamus; (c) in males, urinary retention may develop if the prostate enlarges and compresses the urethra, restricting urine flow.

Related Clinical Terms (Ch 26)

- **Azotemia**: excessive urea or other nitrogen-containing compounds in the blood.
- **Bacteriuria**: abnormal presence of bacteria in the urine.
- **Cystocele**: weakening/stretching of supportive tissue between bladder and vaginal wall, allowing the bladder to bulge into the vagina.
- **Enuresis**: involuntary urination, especially in a sleeping child.
- **Hydroureter**: dilation of a ureter caused by obstruction of urine flow.
- **Nephroptosis**: kidney displaced downward from its usual position ("floating kidney").
- **Nephrotic syndrome**: kidney disorder causing excessive protein excretion in urine.
- **Nephrotoxin**: a toxin with specific harmful effects on the kidney.
- **Polycystic kidney disease**: inherited abnormality affecting development/structure of kidney tubules.
- **Shock-wave lithotripsy**: noninvasive technique to pulverize kidney stones using high-pressure shock waves through a water-filled tub.
- **Urologist**: physician specializing in urinary system function/disorders.

Ch 19.6 — The Adrenal Glands (Endocrine System)

19.6 | The Adrenal Glands

- Each **adrenal gland (suprarenal gland)**: yellow, pyramid-shaped; attached firmly to the superior border of each kidney by a dense fibrous capsule. Nestles among the kidney, diaphragm, and major posterior abdominal wall arteries/veins.
- Adrenal glands project into the peritoneal cavity; anterior surfaces covered by a layer of parietal peritoneum.
- Highly vascularized (like other endocrine glands). Blood supply: branches of the **renal artery**, the **inferior phrenic artery**, and a direct branch from the aorta (the **middle adrenal artery**). **Adrenal veins** carry blood away.
- Weighs ~**7.5 g**; usually heavier in men than women; size varies with secretory demand.
- Each adrenal gland divided into two regions: a superficial **cortex** and an inner **medulla**.

The Adrenal Cortex

- Yellowish color from stored lipids (cholesterol, fatty acids). Produces >**24 different steroid hormones**, called **corticosteroids (adrenocortical hormones)**.
- Deep to the capsule, the cortex has three zones, each synthesizing different steroid hormones. All cortical cells have extensive **smooth endoplasmic reticulum (SER)** for manufacturing lipid-based steroids (contrasts with abundant rough ER in protein-secreting cells like anterior pituitary or thyroid cells).

1. **Zona glomerulosa** (outermost): endocrine cells form densely packed clusters ("glomerulus" = little ball/knot). Occupies ~**15%** of adrenal cortex; extends from capsule to the zona fasciculata.

- ◦ Produces **mineralocorticoids (MCs)** — affect electrolyte composition of body fluids. **Aldosterone** = the main mineralocorticoid: stimulates conservation of sodium ions (Na+) and elimination of potassium ions (K+); causes retention of Na+ by kidneys, sweat glands, salivary glands, and pancreas; prevents Na+ loss in urine, sweat, saliva, digestive secretions. K+ loss accompanies this Na+ retention.
- ◦ Aldosterone secretion triggered by: (1) decreased blood Na+ levels, (2) increased blood K+ levels, or (3) arrival of **angiotensin II** (a hormone produced via the kidney-initiated renin pathway).

2. **Zona fasciculata** (middle): cells have more lipids than zona glomerulosa cells, giving cytoplasm a pale, foamy appearance. Cells form cords radiating outward "like a sunburst" from the zona reticularis; adjacent cords separated by flattened, fenestrated sinusoids (blood vessels).

- ◦ **ACTH** from the anterior pituitary stimulates steroid production here.
- ◦ Produces **glucocorticoids (GCs)** (named for effects on glucose metabolism). Most important: **cortisol (hydrocortisone)** and **corticosterone**. Liver converts some circulating cortisol to **cortisone** (another active glucocorticoid).
- ◦ These hormones speed up rates of glucose synthesis and glycogen formation, especially in the liver. (Table also notes: glucocorticoids release amino acids from skeletal muscle cells — glucose-sparing — and have anti-inflammatory effects.)

3. **Zona reticularis** (deepest/innermost, smallest zone): cells much smaller than adrenal medulla cells, making the cortex-medulla boundary easy to distinguish. Occupies ~**5%** of the adrenal cortex. Cells form a folded, branching network with extensive capillary supply.

- ◦ Secretes small amounts of sex hormones called **androgens**. Adrenal androgens stimulate development of pubic hair in boys and girls before puberty. In women, adrenal androgens promote muscle mass, stimulate blood cell formation, and support sex drive. Less important in men (testes produce androgens in much larger amounts).

The Adrenal Medulla

- Pale gray or pink (due to many blood vessels). Composed of **chromaffin cells —** large, rounded cells resembling sympathetic ganglion neurons. Innervated by preganglionic sympathetic fibers; sympathetic activation via **splanchnic nerves** triggers secretory activity of these modified ganglionic neurons.
- Two secretory cell populations: one secreting **epinephrine (adrenaline)**, the other **norepinephrine (noradrenaline) —** both **catecholamines**. Medulla secretes **three times more epinephrine than norepinephrine**.
- Catecholamine secretion speeds up cellular energy use and mobilization of energy reserves — increases muscular strength and endurance. Metabolic changes peak **30 seconds** after adrenal medulla stimulation and last for several minutes — effects outlast direct sympathetic neural stimulation elsewhere.
- (Summary table): Adrenal medulla hormones (epinephrine, norepinephrine) act on most cells to increase cardiac activity and blood [pressure/glucose availability — table truncated in source].

19.7 | Endocrine Functions of the Kidneys and Heart

- Kidneys produce the enzyme **renin** and two hormones: the peptide **erythropoietin** and the steroid **calcitriol**.
- **Renin-angiotensin pathway**: renin enters circulation → converts circulating **angiotensinogen** (inactive protein produced by the liver) to **angiotensin I** → converted to **angiotensin II** within lung capillaries → angiotensin II stimulates aldosterone secretion by the adrenal cortex.
- **Erythropoietin (EPO)**: peptide hormone released by kidneys in response to low oxygen levels in the kidneys; stimulates red blood cell production in red bone marrow → increases blood volume and improves oxygen delivery to tissues. (Clinical note: EPO is used for "blood doping" by endurance athletes for competitive advantage.)
- **Calcitriol**: steroid hormone secreted by kidneys in response to parathyroid hormone (PTH) levels in the blood. Synthesis depends on availability of **cholecalciferol (vitamin D3)**, synthesized in skin or absorbed from diet. Liver converts cholecalciferol to an intermediary product released into circulation and absorbed by the kidneys; kidneys convert this into calcitriol. Best-known function: stimulates absorption of calcium and phosphate ions

along the digestive tract. Since PTH stimulates calcitriol release, PTH has an indirect effect on intestinal calcium absorption.

- **Atrial natriuretic peptide (ANP)**: produced by cardiac muscle cells in response to increased blood pressure or blood volume. Effects generally oppose those of angiotensin II — promotes loss of Na+ and water by the kidneys, and inhibits renin release, ADH, and aldosterone.

Ch 19.8 — The Pancreas and Other Endocrine Tissues of the Digestive System

19.8 | The Pancreas

- The pancreas, the lining of the digestive tract, and the liver produce exocrine secretions essential for normal digestion. The ANS influences the pace of digestive activities, but most digestive processes are controlled locally by individual organs. Digestive organs communicate with each other using hormones (detailed in Ch 25).
- The pancreas produces hormones that affect metabolic operations throughout the body.
- **Location/gross anatomy**: both exocrine and endocrine gland; lies within the abdominopelvic cavity within the first fold of the small intestine (duodenum), close to the stomach. Slender, pale organ with a lumpy appearance. Adult pancreas ~**20–25 cm (8–10 in.)** long, weighs **80 g (2.8 oz)**. (Detailed gross anatomy covered in Ch 25.)
- **Exocrine pancreas**: makes up ~**99%** of pancreatic volume; produces large quantities of digestive enzyme-rich fluid entering the digestive tract via a network of secretory ducts.
- **Endocrine pancreas**: small groups of endocrine cells scattered throughout the gland, called **pancreatic islets**. Though islets account for only ~**1%** of pancreatic cell population, there are approximately **2 million pancreatic islets** in a normal pancreas.
- Like other endocrine tissues, an extensive fenestrated capillary network surrounds the islets. Blood supply: **pancreaticoduodenal** and **pancreatic arteries**; venous blood returns to the **hepatic portal vein**. Islets innervated by the ANS via branches from the **celiac plexus**.

Cell Types of the Pancreatic Islets (four types):

1. **Alpha cells**: produce **glucagon** — increases blood glucose level by increasing rates of glycogen breakdown and glucose release by the liver.

2. **Beta cells**: produce **insulin** — lowers blood glucose level; increases rate of glucose uptake and utilization by most body cells.

3. **Delta cells**: produce **somatostatin** (growth hormone–inhibiting hormone) — inhibits production/secretion of both glucagon and insulin; slows rates of food absorption and enzyme secretion by the digestive tract.

4. **F cells**: produce **pancreatic polypeptide (PP)** — inhibits gallbladder contractions and regulates production of some pancreatic enzymes; may control rate of nutrient absorption by the digestive tract.

- Alpha and beta cells are sensitive directly to blood glucose concentrations and are NOT under direct control of the nervous system or other endocrine glands: increased blood glucose → beta cells secrete insulin; decreased blood glucose → alpha cells secrete glucagon.

Clinical: Diabetes Mellitus

- Characterized by glucose concentrations high enough to overwhelm the reabsorption capabilities of the kidneys. (Abnormally high blood glucose in general = **hyperglycemia**.) Glucose appears in urine (**glycosuria**), and urine production becomes excessive (**polyuria**).
- Causes: genetic abnormalities (identified genes causing inadequate insulin production, abnormal insulin molecule synthesis, or defective receptor proteins); obesity accelerates onset/severity; can also result from other pathological conditions, injuries, immune disorders, or hormonal imbalances.

- Two major types: **insulin-dependent (Type 1)** and **non-insulin-dependent (Type 2)**. Type 1 controlled (with varying success) via insulin administration (injection or insulin pump infusion). Type 2 most effectively treated with dietary restrictions.
- Because glucose levels can't be adequately stabilized even with treatment, diabetics commonly develop chronic medical problems — tissues experience an "energy crisis," responding as during chronic starvation (breaking down lipids and even proteins because they can't absorb glucose from surroundings).
- Common diabetes-related medical disorders:
  - **Diabetic retinopathy**: proliferation of capillaries and hemorrhaging at the retina, may cause partial or complete blindness.
  - **Cataracts**: changes in lens clarity.
  - **Diabetic nephropathy**: small hemorrhages and inflammation at the kidneys causing degenerative changes leading to kidney failure; the primary cause of kidney failure. Drug treatment improving renal blood flow can slow progression.
  - **Diabetic neuropathy**: peripheral neuropathies and abnormal autonomic function, probably related to disturbances in blood supply to neural tissues.
  - Degenerative changes in cardiac circulation → early heart attacks; heart attacks are **3 to 5 times** more likely in diabetics than nondiabetics of the same age group.
  - Other vascular changes disrupt normal blood flow to distal limbs — e.g., reduced blood flow to the feet can lead to tissue death, ulceration, infection, and loss of toes or a major portion of one or both feet.

Ch 23 — The Spleen (Lymphatic System)

The Spleen

- Largest lymphatic organ in the body; ~**12 cm (5 in.)** long, weighs up to **160 g (5.6 oz)**. Located on the left side, along the lateral border of the stomach, between the **ninth and eleventh ribs**. Attached to the lateral border of the stomach by the **gastrosplenic ligament** (a piece of mesentery).
- Soft; shape reflects surrounding structures. Deep red color on gross dissection due to large blood content.
- Functions (parallels lymph node functions, but for blood): (1) removing abnormal blood cells and other blood components through phagocytosis, (2) storing iron recycled from red blood cell breakdown, (3) initiating immune responses by B cells and T cells in response to antigens in circulating blood.

Surfaces of the Spleen

- **Diaphragmatic surface**: smooth, convex, conforms to the shape of the diaphragm and body wall.
- **Visceral surface** (inferior): contains indentations following the shapes of the stomach (**gastric area**) and left kidney (**renal area**).
- **Hilum**: where the splenic artery, splenic vein, and lymphatic vessels draining the spleen enter/leave.

Histology of the Spleen

- Surrounded by a **capsule** containing collagen and elastic fibers. Cellular components deep to the capsule = the **pulp** of the spleen.
- **Red pulp**: forms **splenic cords**; contains large quantities of red blood cells.
- **White pulp**: forms lymphatic nodules (dominated by lymphocytes).
- Circulatory pathway: **splenic artery** enters at the hilum → branches into numerous **trabecular arteries** radiating outward toward the capsule → these branch extensively; arteriolar branches surrounded by areas of white pulp → capillaries discharge blood into the venous circulation.
- Red pulp cell population includes all normal circulating blood components, plus fixed and free macrophages. A network of **reticular fibers** forms the structural framework of red pulp. Blood passes through this network into large **sinusoids** lined by fixed macrophages. Sinusoids empty into small veins that merge to form **trabecular veins**, continuing toward the hilum.

- This circulatory arrangement lets splenic phagocytes identify and engulf damaged or infected cells in circulating blood. Macrophages are scattered throughout red pulp; the region surrounding white pulp has a high concentration of lymphocytes and dendritic cells — so microorganisms/antigens in the blood quickly trigger an immune response.

Clinical: Lymphoma

- Group of blood cancers developing in the lymphatic system. Begins when a mutation within a lymphocyte (in a lymph node or lymphoid tissue) causes uncontrolled growth and abnormal daughter cells; these accumulate in lymphatic tissues, crowding out normal lymphocytes and impeding lymphatic system function.
- Two main types:
    - **Hodgkin lymphoma**: cancer of blood and bone marrow; ~**10%** of lymphomas. Characterized by an abnormal B cell type called **Reed-Sternberg cells**; associated with Epstein-Barr virus infection. Affects people in their 20s and people over 55; most common in US, Canada, northern Europe; rare in Asia. High concordance in identical twins (if one twin has it, the other has very high risk). Very treatable; most cases curable.
    - **Non-Hodgkin lymphoma (NHL)**: diverse group affecting the lymphatic system, sometimes involving bone marrow and blood; ~**90%** of lymphomas are NHL. ~**85%** of NHL cases affect B cells; remaining involve T cells and NK cells. Described as indolent (slow-growing) or aggressive (fast-growing); can occur at any age but most often affects people over 60. Cause unknown, but certain chemicals (e.g., benzene) and nuclear radiation exposure implicated; also develops more often in immunocompromised individuals.
- Symptoms (both types): enlarged lymph nodes that don't resolve, unexplained weight loss, fever, night sweats, unusual infections.
- Treatment depends on type/stage: chemotherapy, radiation therapy, immunotherapy (using the body's own immune system to attack cancer cells); stem cell transplant considered if these fail.

23.7 | Aging and the Lymphatic System

- With advancing age, T cells become less responsive to antigens → fewer cytotoxic T cells respond to infection. Helper T cell numbers also decline → B cells less responsive, antibody levels don't rise as quickly after antigen exposure.
- Aging therefore increases susceptibility to viral and bacterial infections — hence flu vaccination is strongly recommended for older people.
- Increased cancer incidence in elderly reflects declining immune surveillance, so tumor cells are not eliminated as effectively.

Ch 17 — The Autonomic Nervous System: Collateral Ganglia, Plexuses, and Divisions

Collateral (Prevertebral) Ganglia — Sympathetic Division

- The sympathetic division includes two sympathetic chains (paired, beadlike, flanking the vertebral column), **three collateral ganglia** anterior to the spinal column, and two adrenal medullae.
- **Celiac ganglion**: located at the base of the celiac trunk. Postganglionic fibers innervate the stomach, duodenum, liver, gallbladder, pancreas, spleen, and kidney. Varies in appearance; often a pair of interconnected masses of gray matter.
- **Superior mesenteric ganglion**: located at the base of the superior mesenteric artery. (Postganglionic fiber targets consistent with structures supplied by the superior mesenteric artery — small intestine and proximal large intestine, per artery distribution.)
- **Inferior mesenteric ganglion**: located at the base of the inferior mesenteric artery. Postganglionic fibers innervate the terminal portions of the large intestine, the kidney and bladder, and the sex organs.

- Preganglionic fibers are short (ganglia close to spinal cord); postganglionic fibers are longer, extending a considerable distance to target organs.
- Sympathetic division shows extensive **divergence**: a single preganglionic fiber may innervate up to **32 ganglionic neurons** in several different ganglia — so one CNS sympathetic motor neuron can control a variety of peripheral effectors, producing a complex, coordinated response.
- All preganglionic neurons release **ACh** at synapses with ganglionic neurons. Most postganglionic fibers release **norepinephrine (NE)**; a few release ACh instead (e.g., at neuroeffector junctions in the body wall, skin, and skeletal muscles).

The Adrenal Medulla (as part of the sympathetic pathway)

- Some preganglionic fibers originating between **T5 and T8** pass through the sympathetic chain and the celiac ganglion without synapsing, proceeding directly to the **adrenal medulla**, where they synapse on modified neurons that perform an endocrine function.
- When stimulated, these modified neurons release **epinephrine (E)** and **norepinephrine (NE)** into an extensive capillary network — functioning as hormones affecting other body regions. **Epinephrine (adrenaline)** accounts for **75–80%** of secretory output; the rest is norepinephrine (noradrenaline).
- Circulating blood distributes these hormones throughout the body, altering metabolic activity broadly. Effects resemble sympathetic postganglionic stimulation but differ in two ways: (1) only cells with appropriate receptors are affected by circulating E/NE; (2) effects last much longer than direct sympathetic innervation because the hormones continue diffusing out of circulating blood for an extended period — tissue epinephrine concentrations may remain elevated for up to **30 seconds**, with effects lasting several minutes (compared to just a few seconds for direct varicosity-released neurotransmitter, reabsorbed/broken down/diffused away quickly). This is because the bloodstream lacks the enzymes to break down E/NE, and most tissues have relatively low concentrations of these enzymes.

Sympathetic Activation and Neurotransmitter Release

- ACh released by cholinergic preganglionic neurons during sympathetic activation always stimulates the ganglionic neurons. This leads postganglionic fibers to release NE at neuroeffector junctions (**adrenergic, sympathetic terminals**).
- Postganglionic sympathetic axons form branches resembling a "string of beads" — each bead, or **varicosity**, is packed with mitochondria and neurotransmitter vesicles; varicosities pass along/near the surfaces of many effector cells. A single axon may have **20,000 varicosities**, affecting dozens of surrounding cells. Receptor proteins are scattered across most plasma membranes; there are no specialized postsynaptic membranes (unlike a skeletal neuromuscular junction with a single axon terminal).
- Effects of varicosity-released neurotransmitter last only a few seconds before reabsorption/breakdown/diffusion away.

Plasma Membrane Receptors and Sympathetic Function

- Two classes of sympathetic receptors sensitive to epinephrine/norepinephrine: **alpha receptors** and **beta receptors**. In general, epinephrine stimulates both classes; norepinephrine primarily stimulates alpha receptors. The effector response depends on which receptor type is activated.

Effects of Sympathetic Stimulation

- Reflexive activation of specific effectors (e.g., smooth muscle in skin blood vessels) can occur without engaging the entire division. In a crisis, however, the **entire sympathetic division responds** — termed **sympathetic activation** — affecting peripheral tissues and altering CNS activity; controlled by sympathetic centers in the hypothalamus.
- Effects of sympathetic activation:
  - Increased alertness (via reticular activating system stimulation) — feeling "on edge."
  - Feeling of energy and euphoria, often with disregard for danger and temporary insensitivity to pain.

  - Increased cardiovascular/respiratory center activity (pons, medulla oblongata) → increased heart rate and contraction strength, elevated blood pressure, increased breathing rate/depth.
  - General elevation in muscle tone (via extrapyramidal system stimulation) — appearing tense, possibly shivering.
  - Mobilization of energy reserves: accelerated glycogen breakdown in muscle/liver cells, and lipid release from adipose tissue.

Clinical Notes — Sympathetic Dysfunction

- **Horner syndrome**: sympathetic postganglionic innervation to one side of the face is interrupted (tumor, infection, injury, or brachial plexus trauma). Affected side of the face becomes flushed (decreased vascular tone), no sweating; pupil markedly constricted, eyelid droops, eye appears to retreat into the orbit.
- **Raynaud's disease (phenomenon)**: SNS temporarily causes peripheral vasoconstriction of small arteries in the fingers/toes in response to cold or stress. Cause unknown; affects more women than men. Smoking, caffeine, and vasoconstrictive drugs can worsen it. (Contrast: normal SNS "fight or flight" response constricts superficial blood vessels, reserving blood for muscles/brain in emergencies.)

17.3 | The Parasympathetic Division

- Operates through a series of interconnected neurons. Efferent parasympathetic neurons originate from cranial nerves **III, VII, IX, and X** and sacral spinal nerves **S2–S4**, synapsing with neurons within or near the innervated organ.
- Preganglionic neurons located in the brainstem (mesencephalon, pons, medulla oblongata contain autonomic nuclei associated with cranial nerves III, VII, IX, X) and in sacral spinal segments (autonomic nuclei in spinal segments S2–S4).
- Ganglionic neurons located in peripheral ganglia within or adjacent to target organs — either **terminal ganglia** (near target organs) or **intramural ganglia** (within the tissues of target organs). Preganglionic fibers do not diverge as extensively as sympathetic fibers (typically synapse on 6–8 ganglionic neurons, all in the same ganglion, influencing the same target organ) — so parasympathetic stimulation effects are more specific and localized than sympathetic effects.

Organization and Anatomy of the Parasympathetic Division

- Preganglionic fibers leave the brain in cranial nerves **III (oculomotor), VII (facial), IX (glossopharyngeal), X (vagus)**.
- N III, N VII, N IX control visceral structures in the head; preganglionic fibers synapse in the **ciliary, pterygopalatine, submandibular, and otic ganglia**; short postganglionic fibers continue to peripheral targets.
- **Vagus nerve (X)** provides preganglionic parasympathetic innervation to intramural ganglia within thoracic and abdominopelvic viscera, traveling as far as the last segments of the large intestine. The vagus alone provides roughly **75%** of all parasympathetic outflow.
- Sacral parasympathetic outflow does not join the ventral rami of spinal nerves; instead, preganglionic fibers form distinct **pelvic nerves** that innervate intramural ganglia in the kidney, urinary bladder, terminal large intestine, and sex organs.

General Functions of the Parasympathetic Division (partial list):

- Constriction of the pupils (restricts light entering eyes, aids near focus).
- Secretion by digestive glands: salivary glands, gastric glands, duodenal and other intestinal glands, the pancreas, and the liver.
- Secretion of hormones promoting nutrient absorption by peripheral cells.
- Increased smooth muscle activity along the digestive tract.
- Stimulation/coordination of defecation.
- Contraction of the urinary bladder during urination.
- Constriction of respiratory passageways.
- Reduction in heart rate and force of contraction.

- Sexual arousal and stimulation of sexual glands (both sexes).
- Functions center on relaxation, food processing, and energy absorption. Parasympathetic stimulation increases nutrient content in the blood; cells throughout the body respond by absorbing nutrients for growth and other anabolic activities.
- Neuroeffector junctions are small with narrow synaptic clefts. Effects are short-lived because most released ACh is inactivated by **acetylcholinesterase (AChE)** in the synapse (and any diffusing ACh is deactivated by AChE in surrounding tissue) — so parasympathetic stimulation effects are quite localized and brief.

17.4 | Relationship Between the Sympathetic and Parasympathetic Divisions

- The parasympathetic division uses ACh at all synapses, acting on two receptor types:

1. **Nicotinic receptors**: on surfaces of all ganglionic neurons (both parasympathetic and sympathetic divisions) and at somatic neuromuscular synapses. ACh exposure always causes excitation via opening of chemically gated Na+ channels.

2. **Muscarinic receptors**: found at all cholinergic neuromuscular/neuroglandular junctions in the parasympathetic division, and at the few cholinergic neuroeffector junctions in the sympathetic division. Produce longer-lasting effects than nicotinic receptor stimulation; response (excitatory or inhibitory) reflects activation/inactivation of specific enzymes.

- The sympathetic division has widespread impact reaching visceral organs/tissues throughout the body; the parasympathetic division modifies activity of structures innervated by specific cranial and pelvic nerves (including thoracic/abdominopelvic visceral organs).
- Some organs are innervated by only one division; most vital organs receive **dual innervation** (both sympathetic and parasympathetic), often with **antagonistic (opposite)** effects. Dual innervation is most common in the digestive tract, heart, and lungs — e.g., sympathetic stimulation decreases digestive tract motility, while parasympathetic stimulation increases it.

Anatomy of Dual Innervation

- In the head: parasympathetic postganglionic fibers from the ciliary, pterygopalatine, submandibular, and otic ganglia accompany cranial nerves to peripheral destinations; sympathetic innervation reaches the same structures by traveling directly from the superior cervical ganglia of the sympathetic chain.
- In the thoracic and abdominopelvic cavities, sympathetic postganglionic fibers intermix with parasympathetic preganglionic fibers at a series of **plexuses**: the **cardiac plexus, pulmonary plexus, esophageal plexus, celiac plexus, inferior mesenteric plexus, and hypogastric plexus**. Nerves leaving these plexuses travel with blood vessels and lymphatics supplying visceral organs.
- Autonomic fibers entering the thoracic cavity intersect at the **cardiac plexus** and **pulmonary plexus** — containing sympathetic fibers innervating the heart and parasympathetic fibers innervating heart and lungs.
- The **esophageal plexus** contains descending branches of the vagus nerve and splanchnic nerves leaving the sympathetic chain ganglia on each side.
- Parasympathetic preganglionic fibers of the vagus nerve follow the esophagus into the abdominopelvic cavity, where they join the network of the **celiac plexus (solar plexus)**.
- The **celiac plexus** and the smaller associated **inferior mesenteric plexus** innervate viscera within the abdominal cavity.
- The **hypogastric plexus** contains the parasympathetic outflow of the pelvic nerves, sympathetic postganglionic fibers from the inferior mesenteric ganglion, and sacral splanchnic nerves from the sympathetic chain; it innervates the digestive, urinary, and reproductive organs of the pelvic cavity.

Visceral Reflexes

- Visceral reflexes are autonomic reflexes initiated in the viscera; provide automatic motor responses that can be modified, facilitated, or inhibited by higher centers, especially the hypothalamus.

- Example: a light shone in one eye triggers a visceral reflex (pupillary constriction/dilation), with pupillary nuclei also controlled by hypothalamic centers concerned with emotional states — e.g., pupils constrict when queasy/nauseated, dilate when sexually aroused.
- Each visceral reflex arc = a receptor, a sensory nerve, [and central/motor components — continues beyond the extracted page range].

Ch 22 — Abdominal Aorta and Hepatic Portal System

The Descending Aorta / Abdominal Aorta

- The diaphragm divides the descending aorta into a superior **thoracic aorta** and inferior **abdominal aorta**.
- **Abdominal aorta**: begins immediately inferior to the diaphragm; descends slightly left of the vertebral column, posterior to the peritoneal cavity; often surrounded by a cushion of adipose tissue. At the level of vertebra **L4**, it splits into the right and left **common iliac arteries** — this split point is called the **terminal segment** of the aorta.
- Delivers blood to all abdominopelvic organs and structures. Major visceral branches are **unpaired**, originating on the anterior aortic surface and extending into the mesenteries to reach visceral organs. Branches to the body wall, kidneys, and other extraperitoneal structures are **paired**, originating along the lateral aortic surfaces.

Three Unpaired Visceral Branches of the Abdominal Aorta

1. **Celiac trunk**: supplies liver, stomach, esophagus, gallbladder, duodenum, pancreas, spleen. Divides into three branches:
    - **Left gastric artery**: stomach and inferior esophagus.
    - **Splenic artery**: spleen; also gives rise to the **left gastro-epiploic artery** (stomach) and **pancreatic arteries** (pancreas).
    - **Common hepatic artery**: gives rise to the **hepatic artery proper** (liver), **right gastric artery** (stomach), **cystic artery** (gallbladder), and **gastroduodenal, right gastro-epiploic, and superior pancreaticoduodenal arteries** (duodenal area).

2. **Superior mesenteric artery**: arises ~2.5 cm inferior to the celiac trunk. Supplies pancreas and duodenum (**inferior pancreaticoduodenal artery**), small intestine (**intestinal arteries**), and most of the large intestine (**right colic, middle colic, ileocolic arteries**).

3. **Inferior mesenteric artery**: arises ~5 cm superior to the aorta's terminal segment. Supplies terminal portions of the colon (**left colic and sigmoid arteries**) and rectum (**rectal arteries**).

Five Paired Branches of the Abdominal Aorta

1. **Inferior phrenic arteries**: inferior surface of the diaphragm and inferior esophagus.

2. **Adrenal (suprarenal) arteries**: originate near the base of the superior mesenteric artery; each supplies an adrenal gland.

3. **Renal arteries**: short (~7.5 cm); originate along the posterolateral abdominal aorta, ~2.5 cm inferior to the superior mesenteric artery; travel posterior to the peritoneal lining to reach the adrenal glands and kidneys.

4. **Gonadal arteries**: originate between the superior and inferior mesenteric arteries. In males = **testicular arteries** (long, thin; supply testes/scrotum); in females = **ovarian arteries** (supply ovaries, uterine tubes, uterus). Distribution of gonadal vessels differs by sex.

5. **Lumbar arteries**: small; arise on the posterior aortic surface; supply the vertebrae, spinal cord, and abdominal wall.

Clinical: Aortic Aneurysm

- An aneurysm = a bulge/weakness in an artery wall; can occur anywhere, including aorta, brain, heart. Aortic wall is very elastic (stretches/recoils with blood flow). Classified anatomically as thoracic aortic aneurysm (ascending or descending) or **abdominal aortic aneurysm**.

- Many abdominal aortic aneurysms cause no early symptoms, discovered incidentally on x-ray/ultrasound; some can be palpated/auscultated in thin individuals. Screening recommended for people over 65 with family history, and men over 65 who have ever smoked (one-time ultrasound).
- If ruptured, prognosis very poor — mortality rate >90%. Preventive surgery often recommended for aneurysms >5 cm diameter that are enlarging.
- **Open repair**: large midline incision, patient anesthetized/supine; aorta cross-clamped above/below the aneurysm to stop blood flow; aneurysm sac opened, synthetic tube graft inserted and sutured across the defect; clamps released, aorta wrapped around graft and sutured closed.
- **Endovascular repair** (less invasive, depending on location/branch involvement): small incision into the femoral artery, guide wire passed to the aneurysm site, sheath slid over wire, stent graft inserted through the femoral artery and advanced up the aorta to the aneurysm, then expanded and attached to the internal aortic wall to prevent further expansion.

Arteries of the Pelvis and Lower Limbs (from abdominal aorta terminal branches)

- At vertebra L4, terminal aorta divides into right/left **common iliac arteries** plus the small **median sacral artery** — these carry blood to the pelvis and lower limbs.
- Common iliac arteries travel along the inner ilium surface, descending posterior to the cecum and sigmoid colon. At the lumbosacral joint, each divides into an **internal iliac artery** and **external iliac artery**.
- **Internal iliac artery**: enters the pelvic cavity; supplies the urinary bladder, internal/external pelvic walls, external genitalia, and medial thigh. Major branches: **superior gluteal, internal pudendal, obturator, lateral sacral arteries**. In females, these vessels also supply the uterus and vagina.
- **External iliac artery**: supplies blood to the lower limbs; much larger diameter than the internal iliac. Crosses the iliopsoas, penetrates the abdominal wall between the anterior superior iliac spine and pubic symphysis, and emerges on the anteromedial thigh as the **femoral artery**.

Veins Draining the Abdomen — The Inferior Vena Cava

- Vessels emptying into the **inferior vena cava (IVC)** drain the abdominal wall, gonads, liver, kidneys, adrenal glands, and diaphragm. The **hepatic portal vein** drains the visceral organs within the abdominal cavity (indirectly, via the liver — see below).
- IVC is retroperitoneal; ascends parallel to and right of the aorta. Blood from IVC flows into the right atrium, mixing with superior vena cava blood, then to right ventricle and pulmonary circuit.
- The abdominal portion of the IVC collects blood from **six major veins**:

1. **Lumbar veins**: drain the lumbar abdomen; connected to the azygos vein (right side) and hemi-azygos vein (left side), which empty into the superior vena cava.

2. **Gonadal veins**: right gonadal vein empties directly into the IVC; the left gonadal vein drains into the left renal vein.

3. **Hepatic veins**: leave the liver, empty into the IVC at the level of vertebra T8. (The liver is the only digestive organ draining directly into the IVC.)

4. **Renal veins**: collect blood from the kidneys — the **largest vessels** draining into the IVC.

5. **Adrenal veins**: drain the adrenal glands. Usually only the right adrenal vein drains directly into the IVC; the left drains into the left renal vein.

6. **Phrenic veins**: drain the diaphragm. Only the right phrenic vein drains into the IVC; the left drains into the left renal vein.

The Hepatic Portal System

- Begins in the capillaries of the digestive organs and ends in the liver sinusoids. **The liver is the only digestive organ draining directly into the inferior vena cava** (all other digestive organs drain via the portal system first).

- Blood leaving capillary beds supplied by the **celiac, superior mesenteric, and inferior mesenteric arteries** flows into veins of the hepatic portal system instead of traveling directly to the IVC.
- A blood vessel connecting two capillary beds is called a **portal vessel**; the network is a **portal system**.
- Blood in the hepatic portal system differs from other systemic venous blood because it contains substances absorbed from the stomach and intestines — blood glucose and amino acid levels in the hepatic portal vein often exceed levels found anywhere else in the cardiovascular system. The hepatic portal system delivers these and other absorbed compounds directly to the liver for storage, metabolic conversion, or excretion.
- **Hepatic portal vein**: the largest vessel of the hepatic portal system; delivers venous blood to the liver. Receives blood from three large veins draining organs within the peritoneal cavity:

1. **Inferior mesenteric vein**: collects blood from capillaries in the inferior large intestine. **Left colic vein** and **superior rectal veins** (draining descending colon, sigmoid colon, rectum) empty into it.

2. **Splenic vein**: collects blood from the inferior mesenteric vein, plus veins from the spleen, the lateral border of the stomach (left gastric vein), and the pancreas (pancreatic veins).

3. **Superior mesenteric vein**: collects blood from veins draining the stomach (right gastro-epiploic vein), small intestine (intestinal and pancreaticoduodenal veins), and two-thirds of the large intestine (ileocolic, right colic, middle colic veins).

- The **hepatic portal vein forms through the fusion of the superior mesenteric vein and splenic vein**. The superior mesenteric vein normally contributes the greater volume of blood and most nutrients.
- As the hepatic portal vein proceeds to the liver, it also receives blood from the **gastric veins** (drain the medial border of the stomach) and **cystic veins** (drain the gallbladder).
- Because blood leaving the intestines goes to the liver first, the composition of blood in the systemic circuit remains relatively stable regardless of ongoing digestive activities (the liver buffers/processes absorbed nutrients/toxins before they reach systemic circulation).

Ch 10 — Muscles of the Abdominal Wall

Spinal Flexors: The Oblique and Rectus Muscle Groups

- **Key Point**: The oblique muscles compress underlying structures and rotate the vertebral column. The rectus muscles are important flexors of the vertebral column and are antagonists to the erector spinae muscles.
- The oblique and rectus muscles lie between the vertebral column and the anterior midline. They are divided into three groups: cervical, thoracic, and abdominal. (The oblique/rectus muscles of the trunk and the diaphragm are grouped together because of their common embryological origins.)
- In the thorax, the oblique muscles are the intercostal muscles (external and internal) plus the transversus thoracis. The **abdominal oblique muscles show the same muscular pattern as the thorax**: three layers of muscle, each layer arranged similarly to the thoracic cavity's layering. These abdominal layers are the **external oblique**, **internal oblique** (together called the "abdominal obliques"), and the **transversus abdominis**. This three-layer arrangement strengthens the abdominal wall.

The Rectus Abdominis

- Originates at the **xiphoid process**, inserts onto the **pubic bone**.
- The **linea alba** ("white line") — a band of fibrous connective tissue — divides the rectus abdominis muscle longitudinally (midline).
- **Transverse tendinous inscriptions**: bands of fibrous connective tissue that divide the rectus abdominis into **four repeated segments** (creating the visible "six-pack" appearance).

Muscle Table — Abdominal Wall Muscles (Origin / Insertion / Action / Innervation)

- **External oblique**: Origin = external and inferior borders of ribs 5–12. Insertion = external oblique aponeurosis extending to the linea alba and iliac crest. Action = compresses abdomen; depresses ribs; flexes, laterally flexes, or rotates the vertebral column **to the opposite side**. Innervation = intercostal nerves 5–12, iliohypogastric, and ilioinguinal nerves.
- **Internal oblique**: Origin = thoracolumbar fascia, inguinal ligament, and iliac crest. Insertion = inferior surfaces of ribs 9–12, costal cartilages 8–10, linea alba, and pubis. Action = same as external oblique (compresses abdomen, depresses ribs, flexes/laterally flexes vertebral column), but rotates the vertebral column **to the same side**. Innervation = intercostal nerves 5–12, iliohypogastric, and ilioinguinal nerves.
- **Transversus abdominis**: Origin = cartilages of ribs 7–12, iliac crest. Insertion = linea alba and pubis. Action = compresses the abdomen. Innervation = intercostal nerves 5–12, iliohypogastric, and ilioinguinal nerves.
- **Rectus abdominis**: Origin = superior surface of the pubis around the symphysis. Insertion = cartilages of ribs 5–7 (inferior surfaces) and xiphoid process of sternum. Action = depresses ribs, flexes the vertebral column, and compresses the abdomen. Innervation = intercostal nerves (T7–T12).
- **Diaphragm**: Origin = xiphoid process, ribs 7–12 and associated costal cartilages. Insertion = central tendon sheet. Action = contraction expands the thoracic cavity and compresses the abdominopelvic cavity. Innervation = phrenic nerves (C3–C5).
- (For reference, the related thoracic oblique-group muscles in the same table: scalenus anterior/middle/posterior elevate ribs/flex neck via cervical spinal nerves; external intercostals elevate ribs via intercostal nerves; internal intercostals depress ribs via intercostal nerves; transversus thoracis depresses ribs via intercostal nerves.)

The Diaphragm

- **Key Point**: The diaphragm is a dome-shaped sheet of skeletal muscle separating the thoracic and abdominal cavities. Contracting the diaphragm flattens the muscle, increasing thoracic cavity volume and causing inhalation.
- Major muscle of breathing. Contraction increases thoracic cavity volume, aiding inhalation. Relaxation decreases thoracic cavity volume, aiding exhalation. (Muscles of breathing examined in detail in Ch 24.)
- Serves as the structural/functional boundary between the thoracic and abdominopelvic cavities — relevant to abdominal cavity anatomy as its superior boundary.

Note on scope

- The extracted page range (273–277) also includes cervical/thoracic spinal flexor muscles (scalenes, intercostals — neck/thorax anatomy) and Section 10.5 (Muscles of the Perineal Region and Pelvic Diaphragm — urogenital/anal triangle muscles, levator ani group, etc.), which belong to the neck/thorax and pelvis units respectively rather than the abdominal wall proper, so they are not detailed here (see Thorax and Pelvis study notes for that content).

Ch 12 — Cross-Sectional Anatomy (Vertebral Levels T10, T12, L5)

- These figures show transverse (cross-sectional) views of the trunk at specific vertebral levels, illustrating the spatial relationships of organs, muscles, and vessels at each level. (A cross section at vertebra T8 is also included in the textbook figure sequence, at the thoracic/heart level, for reference/comparison.)

Cross Section at the Level of Vertebra T8 (for reference — thoracic/cardiac level)

- Anterior-to-posterior structures identified: body of sternum, pectoralis major, right atrium, right AV (tricuspid) valve, right ventricle, interventricular septum, left ventricle, right lung (middle lobe, with oblique fissure), left lung (superior lobe, inferior lobe, with oblique fissure), esophagus, thoracic aorta, spinal cord, ribs 4, 7, and 8, spinous process of T8, latissimus dorsi, trapezius.

Cross Section at the Level of Vertebra T10

- Anterior structures: xiphoid process, cardiac orifice of the stomach, cardia of the stomach, right lobe of the liver, esophagus, inferior vena cava, azygos vein, thoracic aorta.
- Other structures at this level: spleen, diaphragm, left lung (inferior lobe), right lung (inferior lobe), body of vertebra T10.
- Posterior structures: latissimus dorsi, multifidus, longissimus thoracis, trapezius; sacral segments of the spinal cord are noted at this level (referring to the spinal cord's internal segmental organization, since the spinal cord itself terminates well above the sacral vertebral level).
- This level captures the transition zone where thoracic organs (esophagus, lower lungs, thoracic aorta) and the most superior abdominal organs (stomach cardia, liver, spleen, diaphragm) are visible together.

Cross Section at the Level of Vertebra T12

- Anterior/abdominal structures: transverse colon, rectus abdominis, transversus abdominis, jejunum, costal cartilage, intercostal muscles of rib 8, rib 9, ascending colon, right lobe of the liver, descending colon, abdominal aorta.
- Renal structures: renal pelvis of the right kidney, renal vein, renal artery, left kidney.
- Diaphragm is visible at this level (marking the abdominal cavity boundary).
- Vertebral/posterior structures: T12–L1 intervertebral disc, spinal cord, latissimus dorsi, psoas major, quadratus lumborum, iliocostalis lumborum, spinalis thoracis, longissimus thoracis.
- This level demonstrates the upper abdominal organ arrangement: liver (right lobe), kidneys with their vessels (renal artery, vein, pelvis), major bowel segments (ascending, transverse, descending colon; jejunum), abdominal aorta, and the posterior body wall musculature (psoas major, quadratus lumborum, erector spinae group).

Cross Section at the Level of Vertebra L5

- Anterior/abdominal-pelvic structures: ileum, rectus abdominis, descending colon, external oblique, cecum, internal oblique, transversus abdominis, psoas (major).
- Pelvic/posterior structures: sacrum, sacro-iliac joint, ilium, ala of sacrum, vertebral foramen, spinous process of L5, iliacus, gluteus medius, gluteus maximus, longissimus thoracis.
- This level shows the transition from abdominal cavity to pelvic cavity: terminal ileum and cecum (ileocecal junction region), descending colon, the three abdominal wall muscle layers (external oblique, internal oblique, transversus abdominis) plus rectus abdominis anteriorly, the psoas and iliacus (iliopsoas group) approaching the pelvis, and the sacroiliac joint/sacral ala posteriorly where the vertebral column meets the pelvis.

Note on scope

- This spread (pp. 336–338) is primarily a labeled-diagram (visual) section of the textbook; the surrounding text in this page range is a clinical case about emergency cricothyrotomy (unrelated to abdominal cross-sectional anatomy) and the start of Chapter 13 (Nervous System) learning outcomes — neither is part of the Abdomen unit content and is omitted here.

Ch 1 — Abdominopelvic Regions/Quadrants, Body Cavities, and the Peritoneal Cavity

Anatomical Regions of the Body (Table 1.1 highlights relevant to trunk/abdomen)

- **Thoracis / Thoracic** = chest. **Abdomen / Abdominal** = abdomen. **Pelvis / Pelvic** = pelvis (in general). **Pubis / Pubic** = anterior pelvis. **Inguen / Inguinal** = groin (crease between thigh and trunk). **Lumbus / Lumbar** = lower back. **Gluteus / Gluteal** = buttock.
- (Other body regions noted in the same table for context: Cephalon/Cephalic = head; Cervicis/Cervical = neck; Brachium/Brachial = upper limb segment closest to trunk; Antebrachium/Antebrachial = forearm; Carpus/Carpal = wrist; Manus/Manual = hand; Femur/Femoral = thigh; Patella/Patellar = kneecap; Crus/Crural = leg, knee to ankle; Sura/Sural = calf; Tarsus/Tarsal = ankle; Pes/Pedal = foot; Planta/Plantar = sole of foot.)

Abdominopelvic Quadrants and Regions

- Anatomists/clinicians use two different methods to describe specific areas of the abdominal/pelvic regions.

Method 1 — Abdominopelvic Quadrants

- The abdominopelvic surface is divided into **four sections** using a pair of imaginary lines (one horizontal, one vertical) intersecting at the **umbilicus (navel)**.
- Useful for describing pain and injuries — location of ache/pain helps determine possible cause. E.g., tenderness in the **right lower quadrant (RLQ)** is a symptom of **appendicitis**; tenderness in the **right upper quadrant (RUQ)** may indicate gallbladder or liver problems.
- **Right Upper Quadrant (RUQ)**: right lobe of liver, gallbladder, right kidney, portions of stomach, small and large intestine.
- **Left Upper Quadrant (LUQ)**: left lobe of liver, stomach, pancreas, left kidney, spleen, portions of large intestine.
- **Right Lower Quadrant (RLQ)**: cecum, appendix, portions of small intestine, reproductive organs (right ovary in female, right spermatic cord in male), right ureter.
- **Left Lower Quadrant (LLQ)**: most of small intestine, portions of large intestine, left ureter, reproductive organs (left ovary in female, left spermatic cord in male).

Method 2 — Nine Abdominopelvic Regions (used for more precise description of internal organ location/orientation)

- Top row: **Right hypochondriac region**, **Epigastric region** (center), **Left hypochondriac region**.
- Middle row: **Right lumbar region**, **Umbilical region** (center), **Left lumbar region**.
- Bottom row: **Right inguinal region**, **Hypogastric (pubic) region** (center), **Left inguinal region**.
- There is a known relationship between superficial anatomical landmarks (quadrants/regions) and the underlying organs (e.g., stomach, liver, spleen, gallbladder, large intestine, small intestine, appendix, urinary bladder are positioned relative to these landmarks) — this is why dividing the abdominal/pelvic area into quadrants or regions is clinically useful.

Anatomical Directions (relevant reference terms)

- **Superior**: above, at a higher level (toward the head). E.g., "The head is superior to the knee."
- **Inferior**: below, at a lower level (toward the feet). E.g., "The knee is inferior to the hip."
- **Cranial/Cephalic**: toward the head. E.g., "The cranial, or cephalic, border of the pelvis is superior to the thigh."
- **Caudal**: toward the tail (coccyx in humans). E.g., "The hips are caudal to the waist."
- **Anterior/Ventral**: the front; before (ventral = abdominal side). E.g., "The navel is on the anterior (or ventral) surface of the trunk."
- **Posterior/Dorsal**: the back; behind. E.g., "The scapula is located posterior to the rib cage."

- **Proximal**: toward an attached base. E.g., "The shoulder is proximal to the wrist."
- **Distal**: away from an attached base. E.g., "The fingers are distal to the wrist."
- **Lateral**: away from the midline. **Medial**: toward the midline.
- **Superficial**: at, near, or relatively close to the body surface. E.g., "The skin is superficial to underlying structures."
- **Deep**: toward the interior of the body, farther from the surface. E.g., "The bone of the thigh is deep to the surrounding skeletal muscles."
- Important convention: anatomical directions always use the **anatomical position** as the standard reference point; left/right refer to the subject's left/right, not the observer's. Equivalent term pairs (posterior/dorsal, anterior/ventral) are not mixed with their opposites in a single description (e.g., don't compare "posterior" to "ventral").

Sectional Anatomy (Sectional Planes)

- Three sectional planes:
  - **Frontal (coronal) plane**: parallels the longitudinal axis; extends side to side, dividing the body into anterior and posterior sections.
  - **Sagittal plane**: parallels the longitudinal axis; extends anterior to posterior, dividing the body into left and right sections. A **midsagittal (median sagittal) section** passes along the midline, dividing the body into roughly equal left/right halves. A **parasagittal section** runs parallel to the midsagittal line but misses the midline (unequal left/right portions).
  - **Transverse (horizontal/cross-sectional) plane**: at right angles to the longitudinal axis of the body part being studied; a division along this plane is a transverse/horizontal/cross section.
- **Serial reconstruction**: an important method for studying histological structure and analyzing images from clinical imaging procedures (CT, MRI); sectional views of the same tubular organ (e.g., a bent tube like elbow macaroni, or the small intestine) can look very different (pair of tubes, dumbbell, oval, solid) depending on where the section is taken.
- Convention: when anatomical diagrams/scans show cross-sectional views, sections are presented as though the observer is standing at the feet of a person in the supine position, looking toward the head (inferior view).

Body Cavities

- The human body is not solid; many organs are suspended in internal chambers called **body cavities** that protect and cushion them.
- The **ventral body cavity** contains organs of the respiratory, cardiovascular, digestive, urinary, and reproductive systems. It is subdivided into the **thoracic cavity** and the **abdominopelvic cavity**, separated by the **diaphragm** (a dome-shaped sheet of skeletal muscle).
- Internal organs projecting into these cavities are called **viscera**. Many organs within these cavities change size/shape during function (e.g., stomach swells at meals, heart contracts/expands each beat) — the cavities allow expansion and limited movement while preventing friction.
- (Note: thoracic, abdominal, and pelvic cavities share a common embryological origin. The term "dorsal body cavity," sometimes used for the cranial/vertebral internal chambers, is anatomically/embryologically distinct — bony-defined, not a true body cavity — and is not used in clinical or comparative anatomy.)

The Thoracic Cavity

- Contains organs of the respiratory, cardiovascular, and lymphatic systems, plus the thymus and inferior esophagus. Bounded by the chest muscles/bones and the diaphragm.
- Subdivided into left and right **pleural cavities** (separated by the **mediastinum**). Each pleural cavity contains a lung, lined by a serous membrane called **pleura** — **visceral pleura** covers the lung's outer surface, **parietal pleura** covers the opposing mediastinal surface and inner body wall — reducing friction during lung expansion/recoil.

- The **mediastinum**: connective tissue surrounding/stabilizing/supporting the esophagus, trachea, thymus, and major blood vessels connecting to the heart. Contains the **pericardial cavity** (small chamber surrounding the heart). The serous membrane covering the heart = **pericardium** (parietal layer = outer fibrous sac; visceral layer = inner layer). Analogy: a fist pushed into a balloon — the wrist = base of the heart, the balloon = the pericardium. The pericardial cavity permits the heart's size/shape changes during each beat; the slippery pericardial lining prevents friction with adjacent mediastinal structures.

The Abdominopelvic Cavity

- Divided into: (1) a superior **abdominal cavity**, (2) an inferior **pelvic cavity**, and (3) an internal chamber called the **peritoneal cavity**.
- **Abdominal cavity**: extends from the inferior surface of the diaphragm to an imaginary plane extending from the inferior surface of the lowest spinal vertebra to the anterior/superior margins of the pelvic girdle. Contains the liver, stomach, spleen, kidneys, pancreas, small intestine, and most of the large intestine. These organs project partially or completely into the peritoneal cavity (analogous to how the heart/lungs project into the pericardial/pleural cavities).
- **Pelvic cavity**: the inferior portion of the abdominopelvic cavity, enclosed by the bones of the pelvis. Contains the last segments of the large intestine, the urinary bladder, and reproductive organs — in females: ovaries, uterine tubes, uterus; in males: prostate and seminal glands. The inferior portion of the peritoneal cavity extends into the pelvic cavity; peritoneum covers the uterine tubes, ovaries, and superior uterus in females, and the superior urinary bladder in both sexes.

The Peritoneal Cavity

- The **peritoneum** is the serous membrane involved: the **parietal peritoneum** lines the body wall; a narrow fluid-filled space separates it from the **visceral peritoneum**, which covers the enclosed organs.
- **Mesenteries**: double sheets of peritoneum that suspend organs such as the stomach, small intestine, and portions of the large intestine within the peritoneal cavity. Mesenteries provide blood supply, support, lubrication, and stability while permitting limited movement.

Clinical Note: Pericarditis and Peritonitis

- The suffix *–itis* means "inflammation." **Pericarditis** = inflammation of the pericardium; can be caused by any disease-causing agent or trauma, and can severely restrict heart function.
- **Peritonitis** = inflammation of the peritoneum (the serous membrane lining the abdomen); may be due to bacterial infection, liver failure, kidney failure, or many other causes. Peritonitis affects all the organs within the peritoneal cavity.

Radiological Procedures (Imaging the Abdomen)

- **X-rays**: high-energy radiation penetrating tissue; a beam strikes a photographic plate, with some x-rays absorbed/deflected by tissue (**radiodensity** = resistance to x-ray penetration). In the human body, increasing radiodensity order: air, fat, liver, blood, muscle, bone. Radiodense tissues (e.g., bone) appear white; less dense tissues appear in shades of gray to black. Physicians specializing in these images = **radiologists**.
  - **Barium-contrast x-ray**: barium is very radiodense, so gastric/intestinal lining contours can be seen outlined against the white barium solution — used for imaging the upper digestive tract.
  - **Digital subtraction angiography (DSA)**: monitors blood flow through specific organs (brain, heart, lungs, kidneys); x-rays taken before/after radiopaque dye administration, computer "subtracts" details common to both images, producing a high-contrast image showing dye distribution.
- **CT scans** (formerly CAT, computerized axial tomography): single x-ray source rotates around the body; x-ray beam strikes a sensor monitored by computer; source completes one revolution every few seconds, then moves and repeats; computer reconstructs a three-dimensional structure from the data, typically displayed as a black-and-white (or colorized) sectional view.
  - **Spiral (helical) CT scan**: newer 3D imaging technology; patient advances at a steady pace through the scanner while the x-ray source rotates continuously — faster/more continuous data gathering yields higher-

quality images with less radiation exposure than a standard CT scanner (which collects one slice at a time more slowly).

- **MRI (magnetic resonance imaging)**: surrounds part/all of the body with a magnetic field ~3000 times as strong as Earth's; this field aligns protons within atomic nuclei (like compass needles in Earth's magnetic field). A radio wave of the proper frequency causes protons to absorb energy; when the pulse ends, that energy is released and detected by MRI computers.
- **Ultrasound**: a small transmitter contacting the skin broadcasts a brief, narrow burst of high-frequency sound and picks up echoes reflected by internal structures; a picture (**echogram**) is assembled from the echo pattern. Images lack the clarity of other procedures, but no adverse effects have been attributed to the sound waves, so fetal development can be monitored without significant risk of birth defects.